

COMPLEX NUMBER

JEE EXERCISES

Exercise J-1

BASICS OF COMPLEX NUMBERS

1. The argument of the complex number $\sin \frac{6\pi}{5} + i \left(1 + \cos \frac{6\pi}{5} \right)$ is :

सम्मिश्र संख्या $\sin \frac{6\pi}{5} + i \left(1 + \cos \frac{6\pi}{5} \right)$ का कोणांक है :

- (A) $\frac{6\pi}{5}$ (B) $\frac{5\pi}{6}$ (C) $\frac{9\pi}{10}$ (D) $\frac{2\pi}{5}$

Ans. C

Sol. $z = \sin \frac{6\pi}{5} + i \left(1 + \cos \frac{6\pi}{5} \right)$

$$\Rightarrow z = \sin \frac{\pi}{5} + i \left(1 + \cos \left(\pi + \frac{\pi}{5} \right) \right)$$

$$z = \sin \frac{\pi}{5} + i \left(1 - \cos \frac{\pi}{5} \right)$$

$\arg(z)$ = lies in IInd quadrant

$$\arg(z) = \pi - \tan^{-1} \left(\frac{1 - \cos \frac{\pi}{5}}{\sin \frac{\pi}{5}} \right)$$

$$= \pi - \tan^{-1} \left(\frac{2 \sin^2 \frac{\pi}{5}}{2 \sin \frac{\pi}{10} \cos \frac{\pi}{10}} \right)$$

$$= \pi - \tan^{-1} \left(\tan \frac{\pi}{10} \right)$$

$$= \pi - \frac{\pi}{10}$$

$$= \frac{9\pi}{10}$$

2. Number of complex numbers z satisfying $z^3 = \bar{z}$ is :

सम्मिश्र संख्या z के मानों की संख्या जो $z^3 = \bar{z}$ को सन्तुष्ट करते हैं :

- (A) 1 (B) 2 (C) 4 (D) 5

Ans. D

Sol. $z = 0$; $z = \pm 1$; $z = \pm i$;

$$z^3 = \bar{z} \Rightarrow |z|^3 = |\bar{z}| = |z|$$

note that $z^n = |\bar{z}|$ has $n+2$ solutions

hence $|z| = 0$ or $|z|^2 = 1$

again $z^4 = z \bar{z} = |z|^2 = 1 \Rightarrow z^4 = 1 \Rightarrow$ total number of roots are 5

Note that (a) the equation $z^n = \bar{z}$ will have $(n+2)$ solutions.

(b) all non-zero solutions of $z^3 = \bar{z}$ are $(1, 0)$, $(-1, 0)$, $(0, 1)$, $(0, -1)$ i.e. a square.

3. For $Z_1 = \sqrt[6]{\frac{1-i}{1+i\sqrt{3}}}$; $Z_2 = \sqrt[6]{\frac{1-i}{\sqrt{3}+i}}$; $Z_3 = \sqrt[6]{\frac{1+i}{\sqrt{3}-i}}$ which of the following holds good?

$$Z_1 = \sqrt[6]{\frac{1-i}{1+i\sqrt{3}}} ; Z_2 = \sqrt[6]{\frac{1-i}{\sqrt{3}+i}} ; Z_3 = \sqrt[6]{\frac{1+i}{\sqrt{3}-i}} \text{ के लिए सत्य कथन है?}$$

(A) $\sum |Z_1|^2 = \frac{3}{2}$

(B) $|Z_1|^4 + |Z_2|^4 = |Z_3|^{-8}$

(C) $\sum |Z_1|^3 + |Z_2|^3 = |Z_3|^{-6}$

(D) $|Z_1|^4 + |Z_2|^4 = |Z_3|^8$

Ans. B

Sol. $|z_1| = \left| \frac{1-i}{1+i\sqrt{3}} \right|^{\frac{1}{6}} = \left| \frac{\sqrt{2}}{2} \right|^{\frac{1}{6}} = 2^{-\frac{1}{12}}$

||ly $|z_2| = 2^{-\frac{1}{12}} ; |z_3| = 2^{-\frac{1}{12}}$ hence the result

4. Let z and w are two non zero complex number such that $|z|=|w|$, and $\text{Arg}(z) + \text{Arg}(w) = \pi$ then :

माना दो सम्मिश्र संख्याएँ z एवं w इस प्रकार से हैं कि $|z|=|w|$, तथा $\text{Arg}(z) + \text{Arg}(w) = \pi$ तो :

- (A) $z = w$ (B) $z = \bar{w}$ (C) $\bar{z} = \bar{w}$ (D) $z = -\bar{w}$

Ans. D

Sol. Let $\arg w = \theta_1$

$$w = |w| (\cos \theta_1 + i \sin \theta_1)$$

$$\arg(z) + \arg(w) = \pi$$

$$\arg z = \pi - \theta_1$$

$$Z = |z| (\cos(\pi - \theta_1) + i \sin(\pi - \theta_1)) \therefore |z| = |w|$$

$$z = -|z| (\cos \theta_1 + i \sin \theta_1)$$

$$z = -\bar{w}$$

5. If z and ω are two non-zero complex numbers such that $|z\omega| = 1$, and $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, then $\bar{z}\omega$ is equal to:

यदि z तथा ω दो अशून्य सम्मिश्र संख्याएँ इस प्रकार हैं कि $|z\omega| = 1$, एवं $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, तो $\bar{z}\omega$ बराबर है :

(A) $-i$

(B) 1

(C) -1

(D) i

Ans. A

Sol. $|z\omega| = 1$

$$\bar{z}\omega = \frac{1}{z\bar{\omega}}$$

$$\text{let } \arg z = \theta_1$$

$$w = |w| (\cos \theta_1 + i \sin \theta_1)$$

$$\therefore \arg(x)$$

$$z = |z| \left(\cos \left(\frac{\pi}{2} + \theta_1 \right) + i \sin \left(\frac{\pi}{2} + \theta_1 \right) \right) - \arg |w| = \frac{\pi}{2}$$

$$\bar{z}\omega = \frac{1}{|z||w| (-\sin \theta_1 + i \cos \theta_1) (\cos \theta_1 + i \sin \theta_1)}$$

$$\bar{z}\omega = \frac{1}{|z||w| i}$$

$$\bar{z}\omega = -i$$

6. If $\left(\frac{1+i}{1-i} \right)^x = 1$, then :

(A) $x = 2n + 1$, where n is any positive integer

(B) $x = 4n$, where n is any positive integer

(C) $x = 2n$, where n is any positive integer

(D) $x = 4n + 1$, where n is any positive integer

यदि $\left(\frac{1+i}{1-i}\right)^x = 1$, तो :

(A) $x = 2n + 1$, जहाँ n एक धनात्मक पूर्णांक है

(B) $x = 4n$, जहाँ n एक धनात्मक पूर्णांक है

(C) $x = 2n$, जहाँ n एक धनात्मक पूर्णांक है

(D) $x = 4n + 1$, जहाँ n एक धनात्मक पूर्णांक है

Ans. B

Sol. $\left(\frac{1+i}{1-i}\right)^x = 1$

$$\left(\frac{(1+i)^2}{(1-i)(1+i)}\right)^x = 1$$

$$(i)x = 4$$

$x = 4n$, where x is positive integer.

7. If $z = x - iy$ and $z^{\frac{1}{3}} = p + iq$, then $\frac{\left(\frac{x}{p} + \frac{y}{q}\right)}{(p^2 + q^2)}$ is equal to :

यदि $z = x - iy$ तथा $z^{\frac{1}{3}} = p + iq$, तो $\frac{\left(\frac{x}{p} + \frac{y}{q}\right)}{(p^2 + q^2)}$ बराबर है :

(A) 1

(B) -1

(C) 2

(D) -2

Ans. D

Sol. $z = x - iy$

$$z^{\frac{1}{3}} = p + iq$$

$$z = p^3 - iq^3 + 3ip^2q - 3pq^2$$

By comparsion

$$p^3 - pq^2 = x$$

$$\frac{x}{p} = p^2 - 3q^2$$

$$3p^2q - q^3 = -y$$

$$\frac{y}{q} = q^2 - 3p^2$$

So that $\frac{\frac{x}{p} + \frac{y}{q}}{p^2 + q^2} = \frac{p^2 - 3q^2 + q^2 - 3p^2}{p^2 + q^2} = -2$

8. If $|z + 4| \leq 3$, then the maximum and minimum value of $|z + 1|$ are :

यदि $|z + 4| \leq 3$, तब $|z + 1|$ का अधिकतम तथा न्यूनतम मान है :

- (A) 4, 1 (B) 4, 0 (C) 6, 0 (D) 6, 1

Ans. C

Sol. $|z + 4| \leq 3$

$$-3 \leq z + 4 \leq 3$$

$$-6 \leq z + 1 \leq 0$$

$$0 \leq |z + 1| \leq 6$$

9. Let α, β be real and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct roots on the line $\text{Re}(z) = 1$, then it is necessary that : [AIEEE-2010]

माना α, β वास्तविक हैं तथा z एक सम्मिश्र संख्या है। यदि $z^2 + \alpha z + \beta = 0$ के रेखा $\text{Re}(z) = 1$ पर दो विभिन्न मूल हैं, तो यह अनिवार्य है कि : [AIEEE-2010]

- (A) $|\beta| = 1$ (B) $\beta \in (1, \infty)$ (C) $\beta \in (0, 1)$ (D) $\beta \in (-1, 0)$

Ans. B

Sol. Let $z^2 + \alpha z + \beta = 0$ has $(1 + iy_1)$ and $(1 + iy_2)$

$$\text{so } z_1 z_2 = \beta$$

$$(1 + iy_1)(1 + iy_2) = \beta$$

$$\beta = 1 - y_1 y_2 + i(y_1 + y_2) \quad (\because \beta \text{ is purely real})$$

$$\text{here } y_1 + y_2 = 0$$

$$y_1 = -y_2$$

$$\beta = 1 - y_1 y_2$$

$$\beta = 1 + y_1^2$$

$$\beta > 1$$

$$\Rightarrow \beta \in (1, \infty)$$

10. If z is a complex number of unit modulus and argument θ , then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ equals : [JEE-(Main)2013]

यदि z एक ऐसी सम्मिश्र संख्या है जिसका मापांक एक इकाई है तथा कोणांक θ है, तो कोणांक $\left(\frac{1+z}{1+\bar{z}}\right)$ बराबर है :

[JEE-(Main)2013]

- (A) $-\theta$ (B) $\frac{\pi}{2} - \theta$ (C) θ (D) $\pi - \theta$

Ans. C

Sol. $\bar{z} = \frac{1}{z}$

$$\arg\left(\frac{1+z}{1+\frac{1}{z}}\right) \Rightarrow \arg z \Rightarrow \theta$$

11. A complex number z is said to be unimodular if $|z| = 1$. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1 - 2z_2}{2 - \bar{z}_1 z_2}$

is unimodular and z_2 is not unimodular. Then the point z_1 lies on a : [JEE-(Main)2015]

- (A) straight line parallel to x-axis (B) straight line parallel to y-axis
(C) circle of radius 2 (D) circle of radius $\sqrt{2}$

एक सम्मिश्र संख्या z एकमापांकी कहलाती है यदि $|z| = 1$ है। माना z_1 तथा z_2 ऐसी सम्मिश्र संख्याएँ हैं कि $\frac{z_1 - 2z_2}{2 - \bar{z}_1 z_2}$ एकमापांकी है

तथा z_2 एकमापांकी नहीं है, तो बिन्दु z_1 स्थित है : [JEE-(Main)2015]

- (A) x-अक्ष के समांतर एक रेखा पर (B) y-अक्ष के समांतर एक रेखा पर
(C) 2 त्रिज्या वाले वृत्त पर (D) $\sqrt{2}$ त्रिज्या वाले वृत्त पर

Ans. C

Sol. $\frac{1}{3} = 1$

$$(z_1 - 2z_2)(\bar{z}_1 - 2\bar{z}_2) = (2 - z_1\bar{z}_2)(2 - \bar{z}_1 z_2)$$

$$z_1\bar{z}_1 - 2z_2\bar{z}_1 - 2z_1\bar{z}_2 + 4z_2\bar{z}_2 = 4 - 2z_1\bar{z}_2 + z_1\bar{z}_1 z_2\bar{z}_2$$

$$|z_1|^2 |z_2|^2 - 4|z_2|^2 - |z_1|^2 + 4 = 0$$

$$(|z_1|^2 - 4)(|z_2|^2 - 1) = 0$$

$$|z_1|^2 - 4 = 0$$

$$|z_1| = 2$$

Circle of radius 2

12. A value of θ for which $\frac{2 + 3i \sin \theta}{1 - 2i \sin \theta}$ is purely imaginary, is : [JEE-(Main)2016]

θ का वह एक मान जिसके लिए $\frac{2 + 3i \sin \theta}{1 - 2i \sin \theta}$ पूर्णतः काल्पनिक है, है : [JEE-(Main)2016]

(A) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Ans. B

Sol. $z = \frac{2 + 3i \sin \theta}{1 - 2i \sin \theta}$

$$z + \bar{z} = 0$$

$$\Rightarrow \sin^2 \theta = 1/3$$

$$\sin \theta = \frac{1}{\sqrt{3}}$$

13. Let $z = \cos \theta + i \sin \theta$. Then the value of $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$ at $\theta = 2^\circ$ is : [JEE-2009]

माना $z = \cos \theta + i \sin \theta$. तब $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$ का मान $\theta = 2^\circ$ पर निम्न है : [JEE-2009]

(A) $\frac{1}{\sin 2^\circ}$ (B) $\frac{1}{3 \sin 2^\circ}$ (C) $\frac{1}{2 \sin 2^\circ}$ (D) $\frac{1}{4 \sin 2^\circ}$

Ans. D

Sol. Given that $z = \cos \theta + i \sin \theta = e^{i\theta}$

$$\therefore \sum_{m=1}^{15} \operatorname{Im}(z^{2m-1}) = \sum_{m=1}^{15} \operatorname{Im}(e^{i\theta})^{2m-1} = \sum_{m=1}^{15} \operatorname{Im}(e^{i(2m-1)\theta}) = \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin 29\theta$$

$$= \frac{\sin\left(\frac{\theta + 29\theta}{2}\right) \sin\left(\frac{15 \times 2\theta}{2}\right)}{\sin\left(\frac{2\theta}{2}\right)} = \frac{\sin(15\theta) \sin(15\theta)}{\sin \theta} = \frac{1}{4 \sin 2^\circ}$$

Hindi. दिया गया है $z = \cos \theta + i \sin \theta = e^{i\theta}$

$$\therefore \sum_{m=1}^{15} \operatorname{Im}(z^{2m-1}) = \sum_{m=1}^{15} \operatorname{Im}(e^{i\theta})^{2m-1} = \sum_{m=1}^{15} \operatorname{Im}(e^{i(2m-1)\theta}) = \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin 29\theta$$

$$= \frac{\sin\left(\frac{\theta + 29\theta}{2}\right) \sin\left(\frac{15 \times 2\theta}{2}\right)}{\sin\left(\frac{2\theta}{2}\right)} = \frac{\sin(15\theta) \sin(15\theta)}{\sin \theta} = \frac{1}{4 \sin 2^\circ}$$

14. Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is : [JEE-2009]

माना $z = x + iy$ एक ऐसी सम्मिश्र संख्या है जिसमें x व y पूर्णांक है। उस आयत का क्षेत्रफल जिसके शीर्ष बिन्दु समीकरण $z\bar{z}^3 + \bar{z}z^3 = 350$ के मूल है, निम्न है : [JEE-2009]

- (A) 48 (B) 32 (C) 40 (D) 80

Ans. A

Sol. $z\bar{z}(z^2 + \bar{z}^2) = 350 \Rightarrow 2(x^2 + y^2)(x^2 - y^2) = 350 \Rightarrow (x^2 + y^2)(x^2 - y^2) = 175$

Since $x, y \in \mathbb{I}$, the only possible case which gives integral solution, is

$x^2 + y^2 = 25$ (1)

$x^2 - y^2 = 7$ (2)

From (1) and (2) $x^2 = 16$; $y^2 = 9$

$\Rightarrow x = \pm 4$; $y = \pm 3 \Rightarrow \text{Area} = 48$

GEOMETRICAL INTERPRETATION AND LOCUS

15. The points z_1, z_2, z_3, z_4 in the complex plane are the vertices of a parallelogram taken in order if and only if :

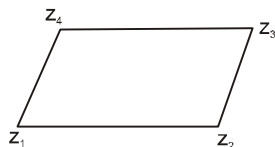
- (A) $z_1 + z_4 = z_2 + z_3$ (B) $z_1 + z_3 = z_2 + z_4$ (C) $z_1 + z_2 = z_3 + z_4$ (D) None

सम्मिश्र समतल में बिन्दु z_1, z_2, z_3, z_4 एक क्रम में लेने पर किसी समान्तर चतुर्भुज के शीर्ष बिन्दु होंगे यदि और केवल यदि :

- (A) $z_1 + z_4 = z_2 + z_3$ (B) $z_1 + z_3 = z_2 + z_4$ (C) $z_1 + z_2 = z_3 + z_4$ (D) कोई नहीं

Ans. B

Sol.



Mid points of diagonals are equal (विकर्णों के मध्य बिन्दु समान होंगे)

$$\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2} \Rightarrow z_1 + z_3 = z_2 + z_4$$

16. The number of solutions of the system of equations $\text{Re}(z^2) = 0, |z| = 2$ is :

समीकरण निकाय $\text{Re}(z^2) = 0, |z| = 2$ के हलों की संख्या है :

- (A) 4 (B) 3 (C) 2 (D) 1

Ans. A

Sol. Let $z = x + iy$

$z^2 = x^2 - y^2 + 2ixy$

$\text{Re}(z^2) = 0$

$$x^2 - y^2 = 0 \Rightarrow x = y \text{ or } x = -y$$

$$|z| = -2$$

$$x^2 + y^2 = 4$$

$$x - y \Rightarrow 2x^2 = 4$$

$$x = +\sqrt{2}, y = +\sqrt{2}$$

$$x = -\sqrt{2}, y = -\sqrt{2}$$

$$x = -y \Rightarrow 2x^2 = 4$$

$$x = \sqrt{2}, y = -\sqrt{2}$$

$$x = -\sqrt{2}, y = \sqrt{2}$$

17. If $z (\neq -1)$ is a complex number such that $\frac{z-1}{z+1}$ is purely imaginary, then $|z|$ is equal to :

यदि $z (\neq -1)$ एक सम्मिश्र संख्या इस प्रकार है कि $\frac{z-1}{z+1}$ विशुद्ध काल्पनिक है, तो $|z|$ का मान है :

- (A) 1 (B) 2 (C) 3 (D) 5

Ans. A

Sol. $\frac{z-1}{z+1} = \lambda i, \lambda \text{ real} \Rightarrow z = \frac{1+i\lambda}{1-\lambda i} \Rightarrow |z| = 1$

18. The curve represented by $|z| = \operatorname{Re}(z) + 2$ is :

- (A) a straight line (B) a circle (C) an ellipse (D) a parabola

$|z| = \operatorname{Re}(z) + 2$ द्वारा प्रदर्शित वक्र है :

- (A) सरल रेखा (B) वृत्त (C) दीर्घवृत्त (D) परवलय

Ans. D

Sol. $|z| = \operatorname{Re}(z) + 2$

Let $z = x + iy$

$$\sqrt{x^2 + y^2} = x + 2$$

$$x^2 + y^2 = (x + 2)^2$$

$$y^2 = 4x + 4$$

Hence the curve is parabola

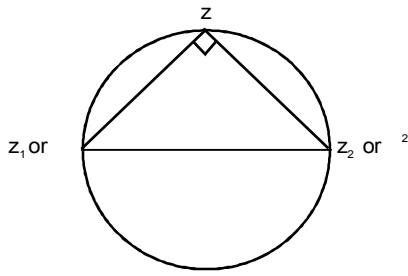
19. Let z_1 and z_2 be two nonreal complex cube roots of unity and $|z - z_1|^2 + |z - z_2|^2 = \lambda$ be the equation of a circle with z_1, z_2 as ends of a diameter then the value of λ is :

माना इकाई के दो अवास्तविक घनमूल z_1 व z_2 हैं तथा $|z - z_1|^2 + |z - z_2|^2 = \lambda$ उस वृत्त को प्रदर्शित करता है जिसके व्यास के सिरे z_1 व z_2 हों, तो λ का मान है :

- (A) 4 (B) 3 (C) 2 (D) $\sqrt{2}$

Ans. B

Sol. $\therefore$ Circle



so by pythagorous theorem $\lambda = |w - w^2|^2 = \left| \sqrt{3} \right|^2 = 3$

20. If z is a complex number satisfying the equation $|z - (1 + i)|^2 = 2$ and $\omega = \frac{2}{z}$, then the locus traced by ' ω ' in the complex plane is :

यदि सम्मिश्र संख्या z समीकरण $|z - (1 + i)|^2 = 2$ और $\omega = \frac{2}{z}$ को सन्तुष्ट करता है तब सम्मिश्र तल में ' ω ' का बिन्दुपथ होगा :

- (A) $x - y - 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y + 1 = 0$ (D) $x + y + 1 = 0$

Ans. A

Sol. We have $|z - (1 + i)|^2 = 2$

$$\Rightarrow (x - 1)^2 + (y - 1)^2 = 2 \quad (\text{Put } z = x + iy)$$

$$\Rightarrow x^2 + y^2 = 2(x + y) \quad \dots\dots(1)$$

$$\text{Let } \omega = h + ik = \frac{2}{z} = \frac{2}{x + iy} = \frac{2(x - iy)}{x^2 + y^2}, \text{ so}$$

$$h = \frac{2x}{x^2 + y^2}, \quad k = \frac{-2y}{x^2 + y^2}$$

$$\Rightarrow h - k = \frac{2(x + y)}{x^2 + y^2} = 1 \quad (\text{from equation (1)})$$

$\therefore$ Locus of the point $\omega(h, k)$ will be $x - y = 1$ Ans. (A)

21. If $|z-2| \geq |z-4|$ then correct statement is :

यदि $|z-2| \geq |z-4|$ तो सही कथन है :

- (A) $\operatorname{Re}(z) \geq 3$ (B) $\operatorname{Re}(z) \leq 3$ (C) $\operatorname{Re}(z) \geq 2$ (D) $\operatorname{Re}(z) \leq 2$

Ans. A

Sol. $|z-2| \geq |z-4|$

Let $z = x + iy$

$$|x + iy - 2| \geq |x + iy - 4|$$

$$|(x-2) + iy| \geq |(x-4) + iy|$$

$$(x-2)^2 + y^2 \geq (x-4)^2 + y^2$$

$$x + 4 - 4x \geq x^2 + 16 - 8x$$

$$x \geq 3$$

22. If $w = \frac{z}{z - \frac{1}{3}i}$ and $|w| = 1$, then z lies on :

- (A) an ellipse (B) a circle (C) a straight line (D) a parabola

यदि $w = \frac{z}{z - \frac{1}{3}i}$ एवं $|w| = 1$, तो z स्थित है :

- (A) एक दीर्घ वृत्त पर (B) एक वृत्त पर (C) एक रेखा पर (D) एक परवलय पर

Ans. C

Sol. $w = \frac{z}{z - \frac{1}{3}i}, |w| = 1$

$$\left| \frac{z}{z - \frac{1}{3}i} \right| = 1$$

$$\frac{|z|}{|z - \frac{1}{3}i|} = 1 \Rightarrow |z| = \left| z - \frac{1}{3}i \right|$$

$$x^2 + y^2 = (x)^2 + \left(y - \frac{1}{3} \right)^2$$

$$6y = 1$$

It's equation of straight line.

23. For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is :

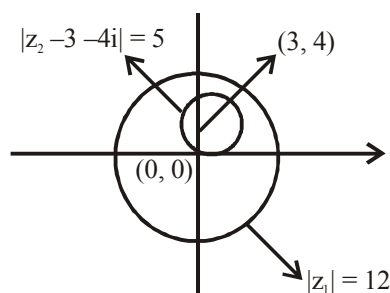
$|z_1| = 12$ और $|z_2 - 3 - 4i| = 5$ को संतुष्ट करने वाली सभी सम्मिश्र संख्याओं z_1, z_2 के लिये $|z_1 - z_2|$ का न्यूनतम मान है :

- (A) 0 (B) 2 (C) 7 (D) 17

Ans. B

Sol. $|z_1| = 12 \Rightarrow$ It is a circle whose centre (0, 0) and radius = 12

$|z_2 - 3 - 4i| = 5 \Rightarrow$ It is a circle whose centre (3, 4) and radius 5



minimum of $|z_1 - z_2| = 12 - 10 = 2$

24. If $Z \neq 1$ and $\frac{z^2}{z-1}$ is real, then the point represented by the complex number z lies : [AIEEE-2012]

- (A) On the imaginary axis
(B) Either on the real axis or on a circle passing through the origin
(C) On a circle with centre at the origin
(D) Either on the real axis or on a circle not passing through the origin

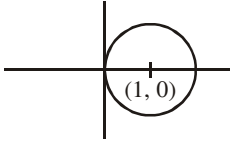
यदि $Z \neq 1$ तथा $\frac{z^2}{z-1}$ वास्तविक है, तो z द्वारा निरूपित सम्मिश्र संख्या वाला बिन्दु : [AIEEE-2012]

- (A) काल्पनिक अक्ष पर है
(B) वास्तविक अक्ष पर है या मूल बिन्दु से होकर जाने वाले वृत्त पर है
(C) एक वृत्त पर है जिसका केन्द्र मूल बिन्दु पर है
(D) वास्तविक अक्ष पर है या एक वृत्त पर है जो मूल बिन्दु से होकर नहीं जाता

Ans. B

Sol. $\frac{z^2}{z-1}$ is purely real where ($Z \neq 1$)

so $\frac{\bar{z}^2}{\bar{z} - 1} = \frac{z^2}{z - 1}$



$$z\bar{z}^2 - \bar{z}^2 = \bar{z}z^2 - z^2$$

$$z\bar{z}(z - \bar{z}) = z^2 - \bar{z}^2$$

$$z\bar{z}(z - \bar{z}) = (z + \bar{z})(z - \bar{z})$$

$$\Rightarrow \bar{z} - z = 0 \text{ or } z + \bar{z} = z\bar{z}$$

$$\Rightarrow \bar{z} = z \text{ or } x^2 + y^2 - 2x = 0$$

$$(x - 1)^2 + y^2 = 1$$

so either lie on z real axis or on a circle passing through the origin.

25. If z is a complex number such that $|z| \geq 2$, then the minimum value of $\left|z + \frac{1}{2}\right|$: [JEE-(Main)-2014]

(A) is strictly greater than $\frac{5}{2}$

(B) is strictly greater than $\frac{3}{2}$ but less than $\frac{5}{2}$

(C) is equal to $\frac{5}{2}$

(D) lies in the interval $(1, 2)$

यदि z एक ऐसी सम्मिश्र संख्या है कि $|z| \geq 2$ है, तो $\left|z + \frac{1}{2}\right|$ का न्यूनतम मान : [JEE-(Main)2014]

(A) $\frac{5}{2}$ से निरंतर बढ़ा है

(B) $\frac{3}{2}$ से निरंतर बढ़ा है परन्तु $\frac{5}{2}$ से कम है

(C) $\frac{5}{2}$ के बराबर है

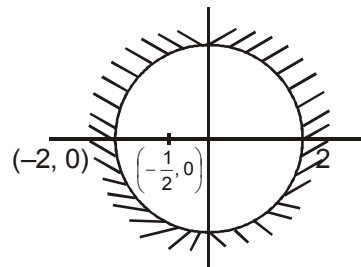
(D) अन्तराल $(1, 2)$ में स्थित है

Ans. D

Sol. $\left|z + \frac{1}{2}\right|$ is distance of 'z' from $-\frac{1}{2}$.

Clearly its minimum value is when $z = -2$ which is $3/2$.

(D) Ans.



26 If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1 - z^2}$ lie on : [JEE-2007]

(A) a line not passing through the origin

(B) $|z| = \sqrt{2}$

(C) the x-axis

(D) the y-axis

यदि $|z| = 1$ तथा $z \neq \pm 1$ तो $\frac{z}{1-z^2}$ के सभी मान स्थित हैं :

[JEE-2007]

(A) एक रेखा जो मूल बिन्दु से नहीं गुजरती है

(B) $|z| = \sqrt{2}$

(C) x-अक्ष

(D) y-अक्ष

Ans. D

Sol. $|z| = 1, z \neq \pm 1$

Let $z = e^{i\theta}$

$$\Rightarrow \frac{z}{1-z^2} = \frac{e^{i\theta}}{1-e^{i2\theta}} = \frac{1}{-2i \sin \theta} \text{ which is purely imaginary}$$

Hence $\frac{z}{1-z^2}$ lies on the y-axis

27. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1 . From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in anticlockwise direction on a circle with centre at origin, to reach a point z_2 . The point z_2 is given by : [JEE-2007]

एक कण P बिन्दु $z_0 = 1 + 2i$ से शुरू होता है, जहाँ $i = \sqrt{-1}$ है। यदि वह पहले मूल बिन्दु से विपरित क्षैतिज दिशा में 5 इकाई तथा इसके बाद मूल बिन्दु से विपरीत उर्ध्वाधर दिशा में 3 इकाई जाता है तो बिन्दु z_1 पर पहुँचता है। z_1 से सदिश $\hat{i} + \hat{j}$ की दिशा में कण $\sqrt{2}$ इकाई दूरी तय करता है। तत्पश्चात् कण मूल बिन्दु को केन्द्र मानते हुये वामावृत्त दिशा में $\frac{\pi}{2}$ से घूर्णन करके z_2 पर पहुँचता है। बिन्दु z_2 है :

[JEE-2007]

(A) $6 + 7i$

(B) $-7 + 6i$

(C) $7 + 6i$

(D) $-6 + 7i$

Ans. D

Sol. $z_0 \equiv (1 + 2i)$

$z_1 \equiv (6 + 5i)$

$z_2 \equiv (-6 + 7i)$

ANGLES AND ROTATION OF COMPLEX NUMBER

28. If z_1, z_2, z_3 are vertices of an equilateral triangle inscribed in the circle $|z| = 2$ and if $z_1 = 1 + i\sqrt{3}$, then :

यदि एक वृत्त $|z| = 2$ के अन्तर्गत बनाए गए समबाहु त्रिभुज के शीर्ष z_1, z_2, z_3 हो तथा $z_1 = 1 + i\sqrt{3}$ हो, तो :

(A) $z_2 = -2, z_3 = 1 + i\sqrt{3}$

(B) $z_2 = 2, z_3 = 1 - i\sqrt{3}$

(C) $z_2 = -2, z_3 = 1 - i\sqrt{3}$

(D) $z_2 = 1 - i\sqrt{3}, z_3 = -1 - i\sqrt{3}$

Ans. C

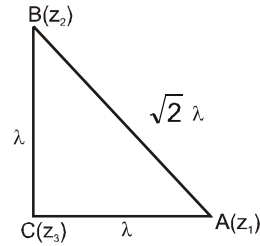
Sol. Apply Rotation (घूर्णन प्रमेय से)

$$\sqrt{2}(z_3 - z_1) = (z_2 - z_1) e^{i\pi/4}$$

$$(z_1 - z_2) = \sqrt{2}(z_3 - z_2) e^{i\pi/4}$$

$$\frac{\sqrt{2}(z_3 - z_1)}{(z_1 - z_2)} = \frac{(z_2 - z_1)}{\sqrt{2}(z_3 - z_2)}$$

$$2(z_3 - z_2)(z_1 - z_3) = (z_1 - z_2)^2$$



29. If z_1 and $\bar{z}_1$ represent adjacent vertices of a regular polygon of n sides with centre at the origin and if $\frac{\text{Im } z_1}{\text{Re } z_1} = \sqrt{2} - 1$ then the value of n is equal to :

यदि z_1 और $\bar{z}_1$ किसी n भुजा वाले समबहुभुज के क्रमागत शीर्ष है जिसका केन्द्र मूल बिन्दु है। यदि $\frac{\text{Im } z_1}{\text{Re } z_1} = \sqrt{2} - 1$ तब n बराबर :

(A) 8

(B) 12

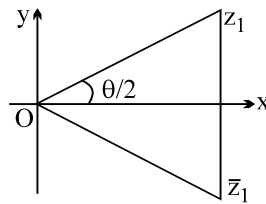
(C) 16

(D) 24

Ans. A

Sol. $\frac{y}{x} = \tan \frac{\theta}{2} = \sqrt{2} - 1 = \tan \frac{\pi}{8}$

$$= \frac{\theta}{2} = \frac{\pi}{8} \Rightarrow \theta = 45^\circ \Rightarrow n = \frac{360^\circ}{45^\circ} = 8$$



if $\frac{y}{x} = 2 - \sqrt{3} \Rightarrow n = 12$

30. Let P denotes a complex number z on the Argand's plane, and Q denotes a complex number $\sqrt{2|z|^2} \text{CiS}\left(\frac{\pi}{4} + \theta\right)$

where $\theta = \text{amp } z$. If 'O' is the origin, then the ΔOPQ is :

(A) isosceles but not right angled

(B) right angled but not isosceles

(C) right isosceles

(D) equilateral

यदि सम्मिश्र तल में P सम्मिश्र संख्या z को निरूपित करता है तथा Q सम्मिश्र संख्या $\sqrt{2}|z|^2 \text{ Cis}\left(\frac{\pi}{4} + \theta\right)$ है जहाँ $\theta = \text{amp } z$. यदि

'O' मूल बिन्दु है तब ΔOPQ है :

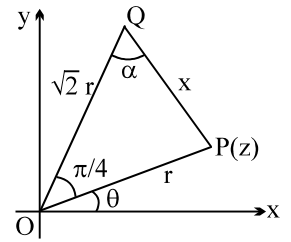
- (A) समद्विबाहु लेकिन समकोण नहीं (B) समकोण लेकिन समद्विबाहु नहीं
(C) समकोण समद्विबाहु (D) समबाहु त्रिभुज

Ans. C

Sol. $Z_p = r \text{ Cis}\theta$; $Z_Q = \sqrt{2}|z| \text{ Cis}\left(\theta + \frac{\pi}{4}\right) = \sqrt{2}r \left[\cos\left(\theta + \frac{\pi}{4}\right) + i \sin\left(\theta + \frac{\pi}{4}\right) \right]$

$$\cos \frac{\pi}{4} = \frac{2r^2 + r^2 - x^2}{2 \cdot \sqrt{2} r \cdot r} = \frac{3r^2 - x^2}{2\sqrt{2} r^2}$$

$$\therefore 1 = \frac{3r^2 - x^2}{2r^2} \Rightarrow r^2 = x^2 \Rightarrow x = r$$



31. z_1 and z_2 are two distinct points in an argand plane. If $a|z_1| = b|z_2|$, (where $a, b \in \mathbb{R}$) then the point $\frac{az_1}{bz_2} + \frac{bz_2}{az_1}$ is a point on the :
- (A) line segment $[-2, 2]$ of the real axis (B) line segment $[-2, 2]$ of the imaginary axis
(C) unit circle $|z| = 1$ (D) the line with $\arg z = \tan^{-1} 2$

यदि z_1 और z_2 सम्मिश्र तल में कोई दो बिन्दु है। यदि $a|z_1| = b|z_2|$, (जहाँ $a, b \in \mathbb{R}$) तब बिन्दु $\frac{az_1}{bz_2} + \frac{bz_2}{az_1}$ कहाँ स्थित होगा :

- (A) वास्तविक अक्ष पर रेखा खण्ड $[-2, 2]$ पर (B) काल्पनिक अक्ष पर रेखा खण्ड $[-2, 2]$ पर
(C) इकाई वृत्त $|z| = 1$ पर (D) रेखा जिसके लिए $\arg z = \tan^{-1} 2$ पर

Ans. A

Sol. Assuming $\arg z_1 = \theta$ and $\arg z_2 = \theta + \alpha$.

$$\frac{az_1}{bz_2} + \frac{bz_2}{az_1} = \frac{a|z_1|e^{i\theta}}{b|z_2|e^{i(\theta+\alpha)}} + \frac{b|z_2|e^{i(\theta+\alpha)}}{a|z_1|e^{i\theta}} = e^{-i\alpha} + e^{i\alpha} = 2 \cos \alpha$$

Alternatively: Let $\alpha = \frac{az_1}{bz_2}$; $\frac{1}{\alpha} = \frac{bz_2}{az_1}$; Also $|\alpha| = \frac{|az_1|}{|bz_2|} = \frac{a|z_1|}{b|z_2|} = 1 \Rightarrow \alpha = \frac{1}{\alpha}$

$$\Rightarrow \alpha + \frac{1}{\alpha} = \alpha + \bar{\alpha} = 2 \text{ Re}(\alpha) = 2 \cos \alpha$$

32. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is :

- (A) of area zero (B) right-angled isosceles
(C) equilateral (D) obtuse – angled isosceles

$\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ को संतुष्ट करने वाली सम्मिश्र संख्याएँ z_1, z_2 और z_3 जिस त्रिभुज के शीर्ष हैं, वह हैं :

- (A) शून्य क्षेत्रफल वाला त्रिभुज (B) समकोण समद्विबाहु त्रिभुज
(C) समबाहु त्रिभुज (D) अधिक कोण समद्विबाहु

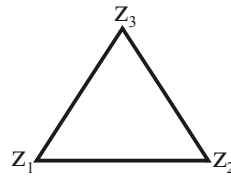
Ans. C

Sol. $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$

$$\arg \left(\frac{z_1 - z_3}{z_2 - z_3} \right) = \left(\frac{-\pi}{3} \right)$$

(-ve sign due to clockwise direction)

$$\left| \frac{z_1 - z_3}{z_2 - z_3} \right| = \left| \frac{1 - i\sqrt{3}}{2} \right|$$



$$\left| \frac{z_1 - z_3}{z_2 - z_3} \right| = 1 \Rightarrow |z_1 - z_3| = |z_2 - z_3|$$

Two sides are equal then corresponding angles are also equal.

Every angle of triangle is $\frac{\pi}{3}$, so that it's an equilateral triangle.

33. The vector $z = -4 + 5i$ is turned counter clockwise through an angle of 180° and stretched 1.5 times. The complex number corresponding to the newly obtained vector is :

- (A) $6 - \frac{15}{2}i$ (B) $-6 + \frac{15}{2}i$ (C) $6 + \frac{15}{2}i$ (D) None of these

सदिश $z = -4 + 5i$ को वामावर्त दिशा में 180° कोण पर घुमाया जाता है तथा 1.5 गुणा विस्तारित कर दिया जाता है। इस प्रकार प्राप्त सदिश के संगत सम्मिश्र संख्या है :

- (A) $6 - \frac{15}{2}i$ (B) $-6 + \frac{15}{2}i$ (C) $6 + \frac{15}{2}i$ (D) इनमें से कोई नहीं

Ans. A

Sol. New $z = \frac{3}{2}(-4 + 5i)e^{i\pi} = 6 - \frac{15}{2}i$

34. Points z_1 and z_2 are adjacent vertices of a regular octagon. The vertex z_3 adjacent to z_2 ($z_3 \neq z_1$) is represented by:

(A) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$

(B) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$

(C) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_2 - z_1)$

(D) None of these

समअष्टभुज के आसन्न शीर्ष z_1 एवं z_2 है, तो z_2 के आसन्न शीर्ष z_3 ($z_3 \neq z_1$) को प्रदर्शित किया जा सकता है :

(A) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$

(B) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$

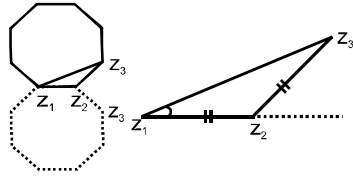
(C) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_2 - z_1)$

(D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{z_3 - z_2}{z_2 - z_1} = e^{\pm i\frac{\pi}{4}}$

$\therefore |z_1 - z_2| = |z_2 - z_3|$



$\Rightarrow z_3 = z_2 + \left(\frac{1 \pm i}{\sqrt{2}}\right)(z_2 - z_1)$

DE MOIVRE'S THEOREM

35. If $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta) \dots (\cos n\theta + i\sin n\theta) = 1$, then the value of θ is :

यदि $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta) \dots (\cos n\theta + i\sin n\theta) = 1$ हो, तो θ का मान है :

(A) $4m\pi, m \in \mathbb{Z}$

(B) $\frac{2m\pi}{n(n+1)}, m \in \mathbb{Z}$

(C) $\frac{4m\pi}{n(n+1)}, m \in \mathbb{Z}$

(D) $\frac{m\pi}{n(n+1)}, m \in \mathbb{Z}$

Ans. C

Sol. $e^{i\theta} \cdot e^{i2\theta} \dots e^{in\theta} = 1 \Rightarrow e^{\frac{i\theta(n)(n+1)}{2}} = e^{i2m\pi} \Rightarrow \frac{\theta n(n+1)}{2} = 2m\pi$

$\theta = \frac{4m\pi}{n(n+1)} \quad m \in \mathbb{Z}.$

36. The complex number ω satisfying the equation $\omega^3 = 8i$ and lying in the second quadrant on the complex plane is :

द्वितीय चतुर्थांश में सम्मिश्र संख्या ω जो समीकरण $\omega^3 = 8i$ को सन्तुष्ट करता है, है :

(A) $-\sqrt{3} + i$ (B) $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$ (C) $-2\sqrt{3} + i$ (D) $-\sqrt{3} + 2i$

Ans. A

Sol. $\omega = 2 \cdot i^{1/3} = 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)^{1/3} = 2 \left[\cos \frac{2n\pi + \frac{\pi}{2}}{3} + i \sin \frac{2n\pi + \frac{\pi}{2}}{3} \right]$

put $n = 1$ $= 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right] = -\sqrt{3} + i$ Ans.

37. If m and n are the smallest positive integers satisfying the relation $\left(2\text{Cis} \frac{\pi}{6} \right)^m = \left(4\text{Cis} \frac{\pi}{4} \right)^n$, then $(m+n)$ has the value equal to :

यदि m और n न्यूनतम धनात्मक पूर्णांक हैं, जो सम्बन्ध $\left(2\text{Cis} \frac{\pi}{6} \right)^m = \left(4\text{Cis} \frac{\pi}{4} \right)^n$ को सन्तुष्ट करते हैं तब :

(A) 120 (B) 96 (C) 72 (D) 60

Ans. C

Sol. $2^{m-2n} \cdot \left[\cos \frac{m\pi}{6} + i \sin \frac{m\pi}{6} \right] = \left[\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right]$

for equality $m = 2n$

$$\frac{n\pi}{4} = \frac{m\pi}{6} + 2k\pi \quad k \in \mathbb{I}$$

put $m = 2n$

$$\frac{n\pi}{4} = \frac{n\pi}{3} + 2k\pi ; \quad -\left(\frac{n\pi}{12} \right) = 2k\pi \quad (\text{ignore } (-)\text{ve sign})$$

$$n = 24k ; \quad m = 48k ; \quad \text{for } m, n \text{ to be smallest } m + n = 72$$

38. Least positive argument of the 4th root of the complex number $2 - i\sqrt{12}$ is :

$2 - i\sqrt{12}$ के चतुर्थ मूल का न्यूनतम धनात्मक कोणांक है :

(A) $\frac{\pi}{6}$

(B) $\frac{5\pi}{12}$

(C) $\frac{7\pi}{12}$

(D) $\frac{11\pi}{12}$

Ans. B

$$\text{Sol. } z^4 = 2(1 - \sqrt{3}i) = 4\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 4\left[\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right]$$

$$z = \sqrt[4]{2}\left[\cos\frac{2m\pi - (\pi/3)}{4} + i\sin\frac{2m\pi - (\pi/3)}{4}\right]$$

$$m = 1, z = \sqrt[4]{2}\left[\cos\left(\frac{5\pi}{12}\right) + i\sin\left(\frac{5\pi}{12}\right)\right]$$

CUBE ROOT OF UNITY

39. If ω is an imaginary cube root of unity then $(1 + \omega - \omega^2)(1 + \omega^2 - \omega)$ equals :

यदि ω इकाई का काल्पनिक घनमूल हो तो $(1 + \omega - \omega^2)(1 + \omega^2 - \omega)$ बराबर है :

(A) 0

(B) 1

(C) 2

(D) 4

Ans. D

$$\text{Sol. } (1 + \omega - \omega^2)(1 + \omega^2 - \omega) \therefore 1 + \omega + \omega^2 = 0$$

$$= (-\omega^2 - \omega^2)(-\omega - \omega)$$

$$= (-2\omega^2)(-2\omega)$$

$$= 4\omega^3$$

$$= 4$$

40. If the cube roots of unity are 1, ω , ω^2 then the roots of the equation $(x - 1)^3 + 8 = 0$, are :

यदि 1, ω तथा ω^2 घनमूल हैं 1 के, तो समीकरण $(x - 1)^3 + 8 = 0$, के मूल हैं :

(A) $-1, -1 + 2\omega, -1 - 2\omega^2$

(B) $-1, -1, -1$

(C) $-1, 1 - 2\omega, 1 - 2\omega^2$

(D) $-1, 1 + 2\omega, 1 + 2\omega^2$

Ans. C

$$\text{Sol. } \text{Since, } (x - 1)^3 + 8 = 0$$

$$\Rightarrow (x - 1)^3 = -8 = (-2)^3 \quad \Rightarrow \left(\frac{x-1}{-2}\right)^3 = 1 \quad \Rightarrow \left(\frac{x-1}{-2}\right) = (1)^{1/3}$$

$$\therefore \text{Cube roots of } \left(\frac{x-1}{-2}\right) \text{ are } 1, \omega \text{ and } \omega^2. \Rightarrow \text{Cube roots of } (x - 1) \text{ are } -2, -2\omega \text{ and } -2\omega^2$$

$$\Rightarrow \text{Cube roots of } x \text{ are } -1, 1 - 2\omega \text{ and } 1 - 2\omega^2.$$

41. If $\omega (\neq 1)$ is a cube root of unity and $(1 + \omega)^7 = A + B\omega$. Then (A, B) equals : [AIEEE-2011]

यदि $\omega (\neq 1)$ इकाई का एक धन मूल है तथा $(1 + \omega)^7 = A + B\omega$ है, तो (A, B) बराबर है : [AIEEE-2011]

- (A) (1, 0) (B) (-1, 1) (C) (0, 1) (D) (1, 1)

Ans. D

Sol. $(1 + \omega)^7 = A + B\omega$

$$(-\omega^2)^7 = A + B\omega$$

$$-\omega^2 = A + B\omega$$

$$1 + \omega = A + B\omega$$

$$A = 1$$

$$B = 1 \quad (1, 1)$$

42. Let ω be a complex number such that $2\omega + 1 = z$ where $z = \sqrt{-3}$. If $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$, then k is equal to :

[JEE-(Main)2017]

माना ω एक सम्मिश्र संख्या ऐसी है कि $2\omega + 1 = z$ जहाँ $z = \sqrt{-3}$ है। यदि $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$ है, तो k बराबर है :

[JEE-(Main)2017]

- (A) -1 (B) 1 (C) -z (D) z

Ans. C

Sol. $x + ky + 3z = 0$ (i)

$$3x + ky - 2z = 0 \quad \dots (ii) \text{ system will have } \infty \text{ solutions}$$

$$2x + 4y - 3z = 0 \quad \dots (iii)$$

$$(i) - (ii)$$

$$x = \frac{5z}{2} \text{ then } y = \frac{-z}{2}$$

$$\frac{xy}{z^2} = 10$$

43. If $\alpha, \beta \in \mathbb{C}$ are the distinct roots, of the equation $x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to :

[JEE-(Main)2018]

यदि $\alpha, \beta \in \mathbb{C}$ समीकरण $x^2 - x + 1 = 0$ के विभिन्न मूल हैं, तो $\alpha^{101} + \beta^{107}$ बराबर है :

[JEE-(Main)2018]

- (A) 1 (B) 2 (C) -1 (D) 0

Ans. A

Sol. Roots of $x^2 - x + 1 = 0$ are $\alpha = -\omega$, $\beta = -\omega^2$
 $\alpha^{101} + \beta^{107} = -\omega^{101} - \omega^{214}$
 $-\omega^2 - \omega = 1$

nth ROOT OF UNITY

44. The value of $\sum_{k=1}^{10} \left(\sin \frac{2k\pi}{11} + i \cos \frac{2k\pi}{11} \right)$ is :

$\sum_{k=1}^{10} \left(\sin \frac{2k\pi}{11} + i \cos \frac{2k\pi}{11} \right)$ का मान है :

(A) 1 (B) -1 (C) -i (D) i

Ans. C

Sol. $\sum_{k=1}^{10} \left(\sin \frac{2k\pi}{11} + i \cos \frac{2k\pi}{11} \right) = i \sum_{k=1}^{10} \left(\cos \frac{2k\pi}{11} - i \sin \frac{2k\pi}{11} \right) = i \left\{ \sum_{k=0}^{10} \left(e^{-\frac{2k\pi}{11}} \right) - 1 \right\}$
 $= -i$

Exercise J-2

BASICS OF COMPLEX NUMBERS

1. Solve the following equations over $\mathbb{C}$ and express the result in the form $a + ib$, $a, b \in \mathbb{R}$.

निम्नलिखित समीकरणों को $\mathbb{C}$ में हल कीजिये तथा अपने परिणाम को $a + ib$, $a, b \in \mathbb{R}$ के रूप में व्यक्त कीजिये।

(a) $ix^2 - 3x - 2i = 0$

(b) $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$

Ans. (a) $-i, -2i$ (b) $\frac{3-5i}{2}$ or $-\frac{1+i}{2}$

Sol. (a) $a + ib$, $a, b \in \mathbb{R}$

$$x = \frac{3 \pm \sqrt{9-8}}{2i}$$

$$x = \frac{3 \pm 1}{2i}$$

$$x = \frac{4}{2i} \times \frac{i}{i} = -2i$$

(b) $(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$

$$x = \frac{4(2-i) \pm \sqrt{16(2i)^2 + 4 \times 2(1+i)(5+3i)}}{4(1+i)}$$

$$x = \frac{4(2-i) \pm \sqrt{16(3i-4i)^2 + 16(1+4i)}}{4(1+i)}$$

$$x = \frac{(2-i) \pm 2}{1+i}$$

$$x = \frac{4-i}{1+i} \times \frac{1-i}{1-i} = \frac{3-5i}{2}$$

$$x = \frac{-i}{1+i} \times \frac{1-i}{1-i} = -\frac{(1+i)}{2}$$

2. Let z be a complex number such that $z \in \mathbb{C} \setminus \mathbb{R}$ and $\frac{1+z+z^2}{1-z+z^2} \in \mathbb{R}$, then prove that $|z|=1$.

माना z एक सम्मिश्र संख्या इस प्रकार है कि $z \in \mathbb{C} \setminus \mathbb{R}$ तथा $\frac{1+z+z^2}{1-z+z^2} \in \mathbb{R}$, तो सिद्ध कीजिये $|z|=1$.

Ans.

Sol. $\frac{1+z+z^2}{1-z+z^2} \in \mathbb{R}$

$$\operatorname{Im} \left(\frac{1+z+z^2}{1-z+z^2} \right) = 0$$

So that

$$\frac{1+z+z^2}{1-z+z^2} = \left(\frac{1+z+z^2}{1-z+z^2} \right)$$

$$(1+z+z^2)(1-\bar{z}+z^2) = (1-z+z^2)(1+z+z^2)$$

$$\Rightarrow 2z - 2\bar{z} = 2(\bar{z}z^2 - z\bar{z}^2)$$

$$= 2(z - \bar{z}) = 2(z|z| - \bar{z}|z|)$$

$$|z| = 1$$

3. Prove the identity, $|1 - z_1 \bar{z}_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$

सर्वसमिका को सिद्ध कीजिये, $|1 - z_1 \bar{z}_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$

Ans.

Sol. $|1 - z_1 \bar{z}_2|^2 - |z_1 - z_2|^2 = (1 - |z_1 \bar{z}_2|)(1 - \bar{z}_1 z_2) - (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$

$$= -\bar{z}_1 z_2 + z_1 \bar{z}_2 + z_1 \bar{z}_1 \bar{z}_2 \bar{z}_2 - z_1 \bar{z}_1 + z_1 \bar{z}_2 + z_2 \bar{z}_1 - \bar{z}_2 \bar{z}_2 + z_2 \bar{z}_1 - z_2 \bar{z}_2$$

$$= 1 - |z_1|^2 |z_2|^2 - |z_1|^2 - |z_2|^2$$

$$= (1 - |z_1|^2) - |z_2|^2 - (1 - |z_1|^2)$$

$$= (1 - |z_1|^2)(1 - |z_2|^2)$$

4. Prove the identity, $|1 + z_1 \bar{z}_2|^2 + |z_1 - z_2|^2 = (1 + |z_1|^2)(1 + |z_2|^2)$

सर्वसमिका को सिद्ध कीजिये, $|1 + z_1 \bar{z}_2|^2 + |z_1 - z_2|^2 = (1 + |z_1|^2)(1 + |z_2|^2)$

Ans.

Sol. $|1 + z_1 \bar{z}_2|^2 + |z_1 - z_2|^2$

$$|1 + z_1 \bar{z}_2|^2 + |z_1 - z_2|^2 = (1 + |z_1|^2)(1 + |z_2|^2)$$

$$= 1 + \bar{z}_1 z_2 + z_1 \bar{z}_2 + z_1 \bar{z}_1 z_2 \bar{z}_2 + z_1 \bar{z}_1 - z_1 \bar{z}_2 - z_2 \bar{z}_1 + z_2 \bar{z}_2$$

$$= 1 + |z_1|^2 |z_2|^2 + |z_1|^2 + |z_2|^2$$

$$= (1 + |z_1|^2) + |z_2|^2 (1 + |z_1|^2)$$

$$= (1 + |z_1|^2) (1 + |z_2|^2)$$

5. (a) Find all non-zero complex numbers Z satisfying $\bar{Z} = i Z^2$.
 (b) If the complex numbers $z_1, z_2, \dots, z_n$ lie on the unit circle $|z| = 1$ then show that

$$|z_1 + z_2 + \dots + z_n| = |z_1^{-1} + z_2^{-1} + \dots + z_n^{-1}|.$$

(a) $\bar{Z} = i z^2$ को संतुष्ट करने वाले सभी अशून्य सम्मिश्र संख्याओं Z को ज्ञात करें।

(b) यदि सम्मिश्र संख्यायें $z_1, z_2, \dots, z_n$ एक ईकाई वृत्त $|z| = 1$ पर स्थित हैं तो सिद्ध कीजिये।

$$|z_1 + z_2 + \dots + z_n| = |z_1^{-1} + z_2^{-1} + \dots + z_n^{-1}|.$$

Ans. (a) $\frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}, i$

Sol. (i) $\bar{Z} = i z^2$

let $z = x + iy$

$$x - iy = i(x + iy)^2$$

$$x - iy = i(x^2 - y^2 + 2ixy)$$

$$x - iy = i(x^2 - y^2) - 2xy$$

By comparison

$$\Rightarrow x = -2xy$$

$$x(2y + 1) = 0$$

$$x = 0, y = \frac{1}{2}$$

$$\Rightarrow x^2 - y^2 = -y$$

$$\text{when } x = 0, y = 0, -1$$

$$\text{and } y = \frac{1}{2}, x = \pm \frac{\sqrt{3}}{2}$$

Non zero, complex number

$$Z = -i, \frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}$$

(B) $|z| = 1$

$$|z|^2 = 1 \Rightarrow z \bar{z} = 1$$

So that $Z = \frac{1}{\bar{Z}}$

$$\begin{aligned} |z_1 + z_2 + z_3 + \dots + z_n| &= |z_1 + z_2 + \dots + z_n| \\ &= |\bar{z}_1 + \bar{z}_2 + \dots + \bar{z}_n| \\ &= \left| \frac{1}{z_1} + \frac{1}{z_2} + \dots + \frac{1}{z_n} \right| \end{aligned}$$

6. Let $z = (0, 1) \in C$. Express $\sum_{k=0}^n z^k$ in terms of the positive integer n .

माना $z = (0, 1) \in C$. $\sum_{k=0}^n z^k$ को धनात्मक पूर्णांक n के रूप में व्यक्त कीजिये।

Ans.
$$\begin{cases} (1, 0) & \text{for } n = 4k \\ (1, 1) & \text{for } n = 4k + 1 \\ (0, 1) & \text{for } n = 4k + 2 \\ (0, 0) & \text{for } n = 4k + 3 \end{cases}$$

Sol. $z = (0, 1) \in C$

$$\sum_{k=0}^n z^k = z^0 + z^1 + z^2 + \dots + z^n$$

7. Let Z is complex satisfying the equation, $z^2 - (3 + i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root, then find the value of m .

माना Z एक सम्मिश्र संख्या है जो समीकरण $z^2 - (3 + i)z + m + 2i = 0$ को संतुष्ट करती है, जहाँ $m \in R$ है। माना समीकरण वास्तविक मूल है तो m का मान ज्ञात कीजियें।

Ans. 2

Sol. $z^2 - (3 + i)z + m + 2i = 0$

Let t be the real root of equation

$$t^2 - (3 + i)t + m + 2i = 0$$

$$t^2 - 3 - it + m + 2i = 0$$

$$(t^2 - 3t + m) + i(-t + 2) = 0$$

Real and Imaginary parts are equal to zero.

$$-t + 2 = 0 \Rightarrow t = +2$$

$$\text{So that } t^2 - 3t + m = 0$$

$$4 - 6 + m = 0$$

$$m = 2$$

8. a, b, c are real numbers in the polynomial, $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$
If two roots of the equation $P(Z) = 0$ are 2 and i, then find the value of 'a'.

a, b, c वास्तविक संख्याये है, बहुपद $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$ है। यदि समीकरण $P(Z) = 0$ के दो मूल 2 तथा i है, तो 'a' का मान ज्ञात करें।

Ans. $-11/2$

Sol. $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$

Roots of the equation are $\alpha = 2, \beta = i, \gamma = -i$ and δ .

$\therefore$ Complex roots appears in conjugate form product of roots $= -\frac{3}{2}$.

$$-t + 2 = 0 \quad t = +2$$

Sum of ro

9. (a) Find the real values of x & y for which $z_1 = 9y^2 - 4 - 10ix$ and $z_2 = 8y^2 - 20i$ are conjugate complex of each other.

वास्तविक संख्यायें x तथा y ज्ञात कीजिये जिनके लिये $z_1 = 9y^2 - 4 - 10ix$ तथा $z_2 = 8y^2 - 20i$ एक दूसरे के सम्मिश्र संयुग्मी है।

- (b) Find the value of $x^4 - x^3 + x^2 + 3x - 5$ if $x = 2 + 3i$

यदि $x = 2 + 3i$ तो $x^4 - x^3 + x^2 + 3x - 5$ का मान ज्ञात कीजिये।

Ans. (a) $[(-2, 2); (-2, -2)]$ (b) $-(77 + 108i)$

Sol. (a) $\bar{z}_1 = z_2 \Rightarrow 9y^2 - 4 + 10ix = 8y^2 - 20i \Rightarrow y^2 = 4$
 $\Rightarrow y = \pm 2 \quad x = -2$

$$(b) \Rightarrow x^4 - x^3 + x^2 + 3x - 5 = (2 + 3i)^4 - (2 + 3i)^3 + (2 + 3i)^2 + 3(2 + 3i) - 5 = -(77 + 108i)$$

10. Solve the following equation $z^2 - (3 - 2i)z = (5i - 5)$ expressing your answer in the form of $(a + ib)$.

समीकरण $z^2 - (3 - 2i)z = (5i - 5)$ को हल करे एवं उत्तर $(a + ib)$ के रूप में व्यक्त करें।

Ans. $z = (2 + i)$ or $(1 - 3i)$

Sol. $z^2 - (3 - 2i)z + (5 - 5i) = 0$

$$z = \frac{(3 - 2i) \pm \sqrt{(3 - 2i)^2 - 4(5 - 5i)}}{2} = \frac{(3 - 2i) \pm \sqrt{-15 + 8i}}{2} = \frac{(3 - 2i) \pm (1 + 4i)}{2}$$

$$= 2 + i \quad \text{or} \quad 1 - 3i$$

11. If $(1 - i)$ is a root of the equation, $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$, then find the other two roots.

यदि समीकरण $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$ का एक मूल $(1 - i)$ है, तो अन्य दो मूल ज्ञात करें।

Ans. $z = 1$ or $2 - i$

Sol. $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$

Let roots of equation are α, β, γ .

$$\alpha = 1 - i$$

Sum of roots

$$\alpha + \beta + \gamma = 2(2 - i)$$

$$1 - i + \beta + \gamma = 4 - 2i$$

$$\beta + \gamma = 3 - i \quad \dots(1)$$

Product of roots

$$\alpha\beta\gamma = 3i - 1$$

$$(1 - i)\beta\gamma = 3i - 1$$

$$\beta\gamma = i - 2 \quad \dots(2)$$

So that other roots are

$$z = 1 \text{ or } 2 - i$$

12. If $iZ^3 + Z^2 - Z + i = 0$, then find $|Z|$.

यदि $iZ^3 + Z^2 - Z + i = 0$ है, तो $|Z|$ ज्ञात करें।

Ans. 1

Sol. $iZ^3 + Z^2 - Z + i = 0$

$$\Rightarrow Z^2 (iZ + 1) + (-Z + i) = 0$$

multiply by i

$$iZ^2 (iZ + 1) + (-iZ - 1) = 0$$

$$(iZ + 1)(iZ^2 - 1) = 0$$

$$\text{Case-I : } Z = -\frac{1}{i} = i \Rightarrow |Z| = 1$$

$$\text{Case II : } Z^2 = \frac{1}{i} \Rightarrow Z^2 = -i$$

$$Z = \frac{1}{\sqrt{2}}i + \frac{1}{\sqrt{2}}i \quad \text{or} \quad -\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \Rightarrow |Z| = 1$$

13. Show that the product, $\left[1 + \left(\frac{1+i}{2}\right)\right] \left[1 + \left(\frac{1+i}{2}\right)^2\right] \left[1 + \left(\frac{1+i}{2}\right)^{2^2}\right] \dots \left[1 + \left(\frac{1+i}{2}\right)^{2^n}\right]$ is equal to $\left(1 - \frac{1}{2^{2^n}}\right) (1+i)$ where $n \geq 2$.

दर्शाइये की गुणन, $\left[1 + \left(\frac{1+i}{2}\right)\right] \left[1 + \left(\frac{1+i}{2}\right)^2\right] \left[1 + \left(\frac{1+i}{2}\right)^{2^2}\right] \dots \left[1 + \left(\frac{1+i}{2}\right)^{2^n}\right]$ का मान $\left(1 - \frac{1}{2^{2^n}}\right) (1+i)$ के बराबर है, जहाँ

$n \geq 2$ है।

Ans.

Sol.
$$\left[1 + \left(\frac{1+i}{2}\right)\right] \left[1 + \left(\frac{1+i}{2}\right)^2\right] \left[1 + \left(\frac{1+i}{2}\right)^{2^2}\right] \dots \left[1 + \left(\frac{1+i}{2}\right)^{2^n}\right]$$

Let $z = \frac{1+i}{2}$

$$= (1+z)(1+z)^2 (1+z)^{2^2} \dots (1+z)^{2^n}$$

multiply and divide by $1-z$

$$= \frac{(1-z^2)}{1-z} (1+z^2)(1+z^{2^2}) \dots (1+z^{2^n})$$

$$\Rightarrow \frac{(1-(z^{2^n})^2)}{1-z}$$

$$\Rightarrow \frac{(1-(z^2)^{2^n})}{1-z}$$

$$= \frac{\left(1 - \left(\frac{1-i}{2}\right)^{2^n}\right)}{1 - \left(\frac{1-i}{2}\right)}$$

$$= \left(1 - \frac{1}{2^{2^n}}\right)(1+i)$$

14. Find the minimum value of the expression $E = |z|^2 + |z-3|^2 + |z-6i|^2$ (where $z = x + iy$, $x, y \in \mathbb{R}$).

चूँकि $E = |z|^2 + |z-3|^2 + |z-6i|^2$ का न्यूनतम मान ज्ञात कीजिये। (जहाँ $z = x + iy$, $x, y \in \mathbb{R}$)

Ans. 30

Sol. $E = |z|^2 + |z-3|^2 + |z-6i|^2$

$$z = x + iy$$

$$E = x^2 + y^2 + x^2 + 9 - 6x + y^2 + x^2 + y^2 + 36 - 12y$$

$$E = 3x^2 - 6x + 3y^2 - 12y + 45$$

$$E = 3(x^2 - 2x + y^2 - 4y + 15)$$

$$E = 3((x-1)^2 + (y-2)^2 + 10)$$

$$E = 3(x-1)^2 + 3(y-2)^2 + 30$$

So that minimum value of E is 30.

15. Let $A = \{a \in \mathbb{R} \mid \text{the equation } (1+2i)x^3 - 2(3+i)x^2 + (5-4i)x + 2a^2 = 0\}$ has at least one real root. Find the value of $\sum_{a \in A} a^2$.

माना $A = \{a \in \mathbb{R} \mid \text{समीकरण } (1+2i)x^3 - 2(3+i)x^2 + (5-4i)x + 2a^2 = 0 \text{ का कम से कम एक वास्तविक मूल है}\}$ $\sum_{a \in A} a^2$ का

मान ज्ञात कीजिये।

Ans. 18

Sol. $(1+2i)x^3 - 2(3+i)x^2 + (5-4i)x + 2a^2 = 0$

Let t is real root of the equation

$$(t^3 - 2t^2 + 5t + 2a^2) + i(2t^3 - 2t^2 - 4t) = 0$$

By comparison

$$\Rightarrow 2t^3 - 2t^2 - 4t = 0 \Rightarrow t = 0, 2, -1$$

$$\text{and } t^3 - 2t^2 + 5t + 2a^2 = 0$$

$$\text{At } t = 0, a^2 = 0 \Rightarrow a = 0$$

$$t = 2, a = \pm\sqrt{3}$$

16. One root of the cubic $2z^3 - (5+6i)z^2 + 9iz + 1 - 3i = 0$ is real. If all the three roots of this cubic are plotted on the complex plane, then find the area of the triangle formed by them.

घन समीकरण $2z^3 - (5+6i)z^2 + 9iz + 1 - 3i = 0$ का एक मूल वास्तविक है। इस घन समीकरणों के मूलों को सम्मिश्र तल में निरूपित करने से बने त्रिभुज का क्षेत्रफल ज्ञात करें।

Ans. $\frac{1}{4}$

Sol. $2z^3 - (5+6i)z^2 + 9iz + 1 - 3i = 0$

$$\text{Put } z = \frac{1}{2}$$

$$\Rightarrow 2\left(\frac{1}{8}\right) - (5+6i)\frac{1}{4} + \frac{9i}{2} + 1 - 3i$$

$$\Rightarrow \frac{1}{4} - \frac{5}{4} - \frac{6i}{4} + \frac{9i}{2} + 1 - 3i$$

$$\Rightarrow 0$$

$$\Rightarrow \left(2 - \frac{1}{2}\right)(2z^2 - 14 + 6i)z - 2 + 6i = 0$$

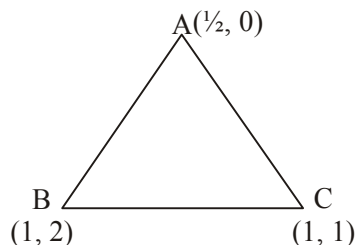
$$\Rightarrow z^2 - (2+3i)z - 1 + 3i = 0$$

$$z = \frac{2+3i \pm \sqrt{4-9+12i+4-12i}}{2 \times 1}$$

$$z = \frac{2+3i \pm i}{2}$$

$$z = 1+2i, 1+i$$

$$\text{Ar}(\Delta ABC) = \frac{1}{2} \begin{vmatrix} \frac{1}{2} & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \left| -\frac{1}{4} \right|$$



$$= \frac{1}{4}$$

17. Prove that $\left(\frac{1+\sin \theta + i \cos \theta}{1+\sin \theta - i \cos \theta} \right)^n = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$. Hence deduce that

$$\left(1 + \sin \frac{\pi}{5} + i \cos \frac{\pi}{5} \right)^5 + i \left(1 + \sin \frac{\pi}{5} - i \cos \frac{\pi}{5} \right)^5 = 0$$

सिद्ध करो कि $\left(\frac{1+\sin \theta + i \cos \theta}{1+\sin \theta - i \cos \theta} \right)^n = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$ तदानुसार

$$\left(1 + \sin \frac{\pi}{5} + i \cos \frac{\pi}{5} \right)^5 + i \left(1 + \sin \frac{\pi}{5} - i \cos \frac{\pi}{5} \right)^5 = 0 \text{ बताइये ।}$$

Ans.

Sol. $\Rightarrow \left(\frac{1+\sin \theta + i \cos \theta}{1+\sin \theta - i \cos \theta} \times \frac{1+\sin \theta + i \cos \theta}{1+\sin \theta + i \cos \theta} \right)^n$

$$\Rightarrow \left(\frac{1+\sin^2 \theta + 2\sin \theta - \cos^2 \theta + 2i \cos \theta (1+\sin \theta)}{1+\sin^2 \theta + 2\sin \theta + \cos^2 \theta} \right)^n$$

$$\Rightarrow \left(\frac{1+\sin^2 \theta + 2\sin \theta - 1 + \sin^2 \theta + 2i \cos \theta (1+\sin \theta)}{2(1+\sin \theta)} \right)^n$$

$$\Rightarrow \left(\frac{2 \sin \theta (1 + \sin \theta) + 2i \cos \theta (1 + \sin \theta)}{2(1 + \sin \theta)} \right)^n$$

$$\Rightarrow (\sin \theta + i \cos \theta)^n$$

$$\Rightarrow \left(\cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right) \right)^n$$

$$\Rightarrow \left(\cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right) \right)$$

18. Dividing $f(z)$ by $z - i$, we get the remainder i and dividing it by $z + i$, we get the remainder $1 + i$. Find the remainder upon the division of $f(z)$ by $z^2 + 1$.

$f(z)$ को $z - i$ से भाग देने पर, हमें शेषफल i प्राप्त होता है तथा इसे $z + i$ से भाग देने पर हमें शेषफल $1 + i$ प्राप्त होता है। $f(z)$ का $z^2 + 1$ से भाग देने पर क्या शेषफल प्राप्त होगा?

Ans. $\frac{iz}{2} + \frac{1}{2} + i$

Sol. Remainder is of form $az + b$ $az + b$ के रूप का शेषफल
 $ai + b = i$
 $-ai + b = 1 + i$

$$b = \frac{1}{2} + i \quad a = \frac{i}{2}$$

$$\text{Remainder} = \frac{i}{2}z + \frac{1}{2} + i \quad \text{शेषफल} = \frac{i}{2}z + \frac{1}{2} + i$$

19. If z_1 and z_2 are two complex numbers such that $|z_1| < 1 < |z_2|$ then prove that $\left| \frac{1 - z_1 \bar{z}_2}{z_1 - z_2} \right| < 1$.

यदि z_1 और z_2 दो सम्मिश्र संख्याएँ इस प्रकार हैं कि $|z_1| < 1 < |z_2|$ हो, तब सिद्ध कीजिए कि $\left| \frac{1 - z_1 \bar{z}_2}{z_1 - z_2} \right| < 1$.

Ans.

Sol. Let $\left| \frac{1 - z_1 \bar{z}_2}{z_1 - z_2} \right| < 1 \Rightarrow |1 - z_1 \bar{z}_2| < |z_1 - z_2|$

$$\Rightarrow (1 - z_1 \bar{z}_2)(1 - \bar{z}_1 z_2) < (z_2 - z_1)(\bar{z}_2 - \bar{z}_1) \Rightarrow 1 + |z_1|^2 |z_2|^2 - |z_1|^2 - |z_2|^2 < 0$$

$$\Rightarrow (1 - |z_1|^2) + (|z_1|^2 - 1) |z_2|^2 < 0 \quad \Rightarrow (1 - |z_1|^2) (1 - |z_2|^2) < 0$$

which is true because of $|z_1| < 1 < |z_2|$.

GEOMETRICAL INTERPRETATION AND LOCUS

20. Among the complex numbers z satisfying the condition $|z + 3 - \sqrt{3}i| = \sqrt{3}$, find the number having the least positive argument.

सम्मिश्र संख्याओं, जो $|z + 3 - \sqrt{3}i| = \sqrt{3}$ को संतुष्ट करती है, में से वह संख्या ज्ञात कीजिये जिसका न्यूनतम धनात्मक कोणांक है।

Ans. $-\frac{3}{2} + \frac{3\sqrt{3}}{2}i$

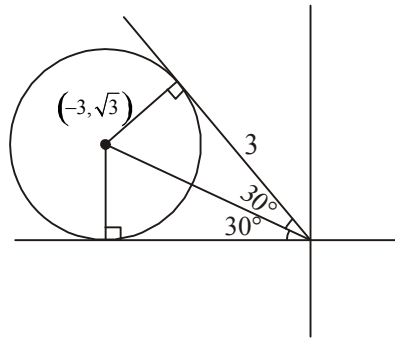
Sol. $12 - (-3 + \sqrt{3}i) = \sqrt{3}$

$$z = 3e^{\frac{i2\pi}{3}}$$

$$z = 3\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$$

$$z = 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$$

$$z = \frac{-3}{2} + \frac{i3\sqrt{3}}{2}$$



21. Let $z_1 = 1 + i$ and $z_2 = -1 - i$. Find $z_3 \in \mathbb{C}$ such that triangle z_1, z_2, z_3 is equilateral.

माना $z_1 = 1 + i$ तथा $z_2 = -1 - i$ है। $z_3 \in \mathbb{C}$ का मान निकालिये जब z_1, z_2, z_3 समबाहु त्रिभुज है।

Ans. $z_3 = \sqrt{3}(1 - i)$ and $z'_3 = \sqrt{3}(-1 + i)$

Sol. Let us have point z_3 as $x + y$

Now, for $\Delta z_1 z_2 z_3$ to be equilateral,

$$z_1 z_2 = z_2 z_3 = z_1 z_3$$

$$\therefore z_1 z_2 = \sqrt{(1 - (-1))^2 + (1 - (-1))^2} = 2\sqrt{2} \quad \text{_____ (1)}$$

$$\therefore z_2 z_3 = \sqrt{(x - (-1))^2 + (y - (-1))^2} \quad \text{_____ (2)}$$

$$\therefore z_1 z_3 = \sqrt{(x+1)^2 + (y+1)^2} \quad \text{_____}(3)$$

equating (1) and (2),

$$2\sqrt{2} = \sqrt{(x-(-1))^2 + (y-(-1))^2}$$

squaring both sides,

$$8 = x^2 + 2x + y^2 + 2y + 2$$

$$\therefore x^2 + 2x + y^2 + 2y - 6 = 0 \quad \text{_____}(4)$$

equating (3) and (1),

$$2\sqrt{2} = \sqrt{(x-1)^2 + (y-1)^2}$$

squaring both sides,

$$8 = x^2 - 2x + y^2 - 2y + 2$$

$$\therefore x^2 - 2x + y^2 - 2y - 6 = 0 \quad \text{_____}(5)$$

subtracting (5) from (4), we have

$$x^2 + 2x + x^2 - 2x - 6 = 0$$

$$\therefore x^2 = 3$$

$$\therefore x = \pm\sqrt{3}$$

Hence, z_3 can be $(\sqrt{3}, -\sqrt{3})$ or $(-\sqrt{3}, \sqrt{3})$.

22. Find the Cartesian equation of the locus of 'z' in the complex plane satisfying, $|z-4| + |z+4| = 16$.

z के बिन्दु पथ का कार्तीय समीकरण ज्ञात करें जो सम्मिश्र तल में $|z-4| + |z+4| = 16$ को संतुष्ट करता है।

Ans. $\frac{x^2}{64} + \frac{y^2}{48} = 1$

Sol. $|z-4| + |z+4| = 16$

$$|z-4| = 16 - |z+4|$$

$$(|z-4|)^2 = (16 - |z+4|)^2$$

$$z = x + y$$

$$(x - y)^2 + y^2 = 16^2 + (x + 4)^2 + y^2 - 32\sqrt{(x + 4)^2 + y^2}$$

$$x^2 + 16 - 8x = 16^2 + x^2 + 16 + 8x - 32\sqrt{(x + 4)^2 + y^2}$$

$$x^2 + 16 - 8x = 16^2 + x^2 + 16 + 8x - 32\sqrt{(x + 4)^2 + y^2}$$

$$32\sqrt{(x + 4)^2 + y^2} = 16x + 16$$

$$2\sqrt{(x + y)^2 + y^2} = x + 16$$

$$(x + 4)^2 + y^2 = \left(\frac{y}{2} + 8\right)^2$$

$$x^2 + 16 + 8x + y^2 = \frac{x^2}{4} + 64 + 8x$$

$$\frac{3x^2}{4} + y^2 = 48$$

23. Interpret the following locii in $z \in \mathbb{C}$:

$z \in \mathbb{C}$ में निम्नलिखित बिन्दु पथों का विवेचना कीजिये :

(a) $1 < |z - 2i| < 3$

(b) $\operatorname{Re} \left(\frac{z + 2i}{iz + 2} \right) \leq 4 \quad (z \neq 2i)$

(c) $\operatorname{Arg}(z + i) - \operatorname{Arg}(z - i) = \pi/2$

(d) $\operatorname{Arg}(z - a) = \pi/3$ where $a = 3 + 4i$

$z \in \mathbb{C}$ में निम्नलिखित बिन्दु पथों का विवेचना कीजिये :

(a) $1 < |z - 2i| < 3$

(b) $\operatorname{Re} \left(\frac{z + 2i}{iz + 2} \right) \leq 4 \quad (z \neq 2i)$

(c) कोणांक $(z + i) - \text{कोणांक}(z - i) = \pi/2$

(d) कोणांक $(z - a) = \pi/3$ जहाँ $a = 3 + 4i$

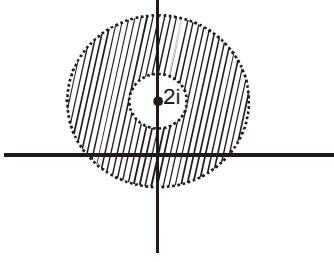
Ans. (a) The region between the co centric circles with centre at $(0, 2)$ and radii 1 and 3 units

(b) region outside or on the circle with centre $\frac{1}{2} + 2i$ and radius $\frac{1}{2}$.

(c) semi circle (in the 1st & 4th quadrant) $x^2 + y^2 = 1$

(d) a ray emanating from the point $(3 + 4i)$ directed away from the origin and having equation $\sqrt{3}x - y + 4 - 3\sqrt{3} = 0$

Sol.



24. Let $A \equiv z_1$; $B \equiv z_2$; $C \equiv z_3$ are three complex numbers denoting the vertices of an acute angled triangle. If the origin 'O' is the orthocentre of the triangle, then prove that :

$$z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 = z_3 \bar{z}_1 + \bar{z}_3 z_1$$

hence show that the ΔABC is a right angled triangle $\Leftrightarrow z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 = z_3 \bar{z}_1 + \bar{z}_3 z_1 = 0$

माना $A \equiv z_1$; $B \equiv z_2$; $C \equiv z_3$ तीन सम्मिश्र संख्यायें एक न्यूनकोण त्रिभुज के शीर्ष हैं। यदि 'O' मूल बिन्दु त्रिभुज का लम्बकेन्द्र है तो सिद्ध कीजिये :

$$z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 = z_3 \bar{z}_1 + \bar{z}_3 z_1$$

तदनुसार दर्शाइये कि त्रिभुज ABC समकोणीय होगा $\Leftrightarrow z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 = z_3 \bar{z}_1 + \bar{z}_3 z_1 = 0$

Ans.

Sol. Here O is orthocenter, then $OA \perp BC$

$$\therefore \arg\left(\frac{z_1 - 0}{z_2 - z_3}\right) = \frac{\pi}{2}$$

$$\therefore \frac{z_1 - 0}{z_2 - z_3} \text{ is purely imaginary.}$$

$$\Rightarrow \frac{z_1 - 0}{z_2 - z_3} + \left(\frac{z_1 - 0}{z_2 - z_3}\right) = 0$$

$$\Rightarrow \frac{z_1}{z_2 - z_3} + \frac{\bar{z}_1}{\bar{z}_2 - \bar{z}_3} = 0$$

$$\Rightarrow z_1 (\bar{z}_2 - \bar{z}_3) + \bar{z}_1 (z_2 - z_3) = 0$$

$$\Rightarrow z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_1 \bar{z}_3 + \bar{z}_1 z_3 \quad \text{_____ (1)}$$

Similarly, $OB \perp AC$

$$\Rightarrow z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 \quad \text{_____ (2)}$$

From (1) and (2),

$$z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 = z_1 \bar{z}_3 + \bar{z}_1 z_3$$

25. Let $z = x + iy$ be a complex number, where x and y are real numbers. Let A and B be the sets defined by $A = \{z \mid |z| \leq 2\}$ and $B = \{z \mid (1-i)z + (1+i)\bar{z} \geq 4\}$. Find the area of the region $A \cap B$.

माना $z = x + iy$ एक सम्मिश्र संख्या है जहाँ x तथा y वास्तविक है। माना A तथा B समुच्चय द्वारा परिभाषित है $A = \{z \mid |z| \leq 2\}$ तथा $B = \{z \mid (1-i)z + (1+i)\bar{z} \geq 4\}$. क्षेत्र $A \cap B$ द्वारा दर्शाया गये क्षेत्र का क्षेत्रफल ज्ञात कीजिये।

Ans. $\pi - 2$

Sol. $z = x + iy$

$$A = \{z \mid |z| \leq 2\}$$

$$\Rightarrow \sqrt{x^2 + y^2} \leq 2$$

$$\text{or } x^2 + y^2 \leq 4$$

$\Rightarrow z$ lies on or inside the circle

$$x^2 + y^2 = 4$$

$$B = \{z : (1-i)z + (1+i)\bar{z} \geq 4\}$$

$$\Rightarrow (1-i)(x+iy) + (1+i)(x-iy) \geq 4$$

$$\Rightarrow x + iy - ix + y + x - iy + ix + y \geq 4$$

$$\Rightarrow x + y \geq 2$$

Area of region $A \cap B$

$$\text{Area} = \frac{\pi(2)^2}{4} - \frac{1}{2} \times 2 \times 2 = \pi - 2$$

26. For all real numbers x , let the mapping $f(x) = \frac{1}{x-i}$, where $i = \sqrt{-1}$. If there exist real number a, b, c and d for which $f(a), f(b), f(c)$ and $f(d)$ form a square on the complex plane. Find the area of the square.

सभी वास्तविक संख्याओं x के लिये माना प्रतिचित्रण $f(x) = \frac{1}{x-i}$, जहाँ $i = \sqrt{-1}$. यदि वास्तविक संख्या a, b, c तथा d जिनके लिये $f(a), f(b), f(c)$ तथा $f(d)$ सम्मिश्र तल में एक वर्ग का निर्माण करते हैं। वर्ग का क्षेत्रफल ज्ञात करो।

Ans. $1/2$

Sol. $f(t) = \frac{1}{t-i} \times \frac{t+i}{t+i}$

also let $f(t) = x + iy$

$$x + iy = \frac{t+i}{t^2+1}$$

$$x = \frac{t}{t^2+1}, y = \frac{1}{t^2+1} \Rightarrow \frac{x}{y} = t$$

$$\text{from } y = \frac{1}{\left(\frac{x}{y}\right)^2 + 1}$$

$$y = \frac{y^2}{x^2 + y^2} \Rightarrow 1 = \frac{y}{x^2 + y^2}$$

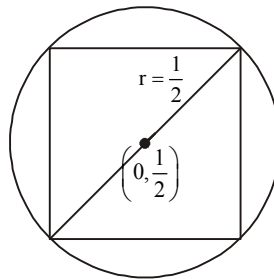
$$x^2 + y^2 - y = 0$$

$$\text{centre} = \left(0, \frac{1}{2}\right)$$

$$\text{radius} = \frac{1}{2}$$

$$\text{So the side of square} = \frac{1}{\sqrt{2}}$$

$$\text{area of square} = \frac{1}{2}$$



27. Find centre and radius of the circle determined by all complex numbers $z = x + iy$ satisfying $\left| \frac{z - \alpha}{z - \beta} \right| = k$, where $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$ are fixed complex and $k \neq 1$.

$\left| \frac{z - \alpha}{z - \beta} \right| = k$, $k \neq 1$ जहाँ $z = x + iy$, $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$ द्वारा दिए जाने वाले वृत्त का केन्द्र ओर त्रिज्या का मान ज्ञात कीजिये।

Ans. Centre $\equiv \frac{k^2\beta - \alpha}{k^2 - 1}$, Radius $= \frac{1}{(k^2 - 1)} \sqrt{|\alpha - k^2\beta|^2 - (k^2 \cdot |\beta|^2 - |\alpha|^2)(k^2 - 1)}$

Sol. As we know $|z|^2 = z \cdot \bar{z}$

$$\text{Given, } \frac{|z - \alpha|^2}{|z - \beta|^2} = k^2$$

$$\Rightarrow (z - \alpha)(\bar{z} - \bar{\alpha}) = k^2 (z - \beta)(\bar{z} - \bar{\beta})$$

$$\Rightarrow |z|^2 - \alpha\bar{z} - \bar{\alpha}z + |\alpha|^2 = k^2 (|z|^2 - \beta\bar{z} - \bar{\beta}z + |\beta|^2)$$

$$\Rightarrow |z|^2 (1 - k^2) - (\alpha - k^2\beta)\bar{z} - (\bar{\alpha} - \bar{\beta}k^2)z + (|\alpha|^2 - k^2|\beta|^2) = 0$$

$$\Rightarrow |z|^2 - \frac{(\alpha - k^2\beta)}{(1 - k^2)}\bar{z} - \frac{(\bar{\alpha} - \bar{\beta}k^2)}{(1 - k^2)}z$$

$$\Rightarrow |z|^2 - \frac{(\alpha - k^2\beta)}{(1 - k^2)}\bar{z} - \frac{(\bar{\alpha} - \bar{\beta}k^2)}{(1 - k^2)}z + \frac{|\alpha|^2 - k^2|\beta|^2}{(1 - k^2)} = 0 \quad \text{---(1)}$$

On comparing with equation of circle, $|z|^2 + a\bar{z} + \bar{a}z + b = 0$

whose centre is $(-a)$ and radius $= \sqrt{|a|^2 - b}$

$\therefore$ Centre for eq. (1)

$$= \frac{\alpha - k^2\beta}{1 - k^2} \text{ and radius}$$

$$= \sqrt{\left(\frac{\alpha - k^2\beta}{1 - k^2}\right)\left(\frac{\bar{\alpha} - \bar{\beta}k^2}{1 - k^2}\right) - \frac{\alpha\bar{\alpha} - k^2\beta\bar{\beta}}{1 - k^2}}$$

$$\text{radius} = \left| \frac{k(\alpha - \beta)}{1 - k^2} \right|$$

28. If one of the vertices of the square circumscribing the circle $|z - 1| = \sqrt{2}$ is $2 + \sqrt{3}i$. Find the other vertices of square.

एक वृत्त $|z - 1| = \sqrt{2}$ के परिगट वर्ग का एक शीर्ष $2 + \sqrt{3}i$ है। वर्ग के अन्य शीर्ष ज्ञात कीजिये।

Ans. $z_2 = -\sqrt{3}i$; $z_3 = (1 - \sqrt{3}) + i$; $z_4 = (1 + \sqrt{3}) - i$

Sol. $z_0 = 1$

$$z_1 = 2 + \sqrt{3}i$$

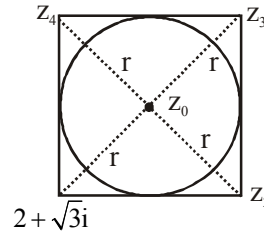
$$z_2 - z_0 = (2 + \sqrt{3}i - z_0)e^{i\pi/2}$$

$$\Rightarrow z_2 = 1 + (1 + \sqrt{3}i)e^{i\pi/2} = (1 - \sqrt{3}) + i$$

Similarly

$$z_3 = 1 + (1 + \sqrt{3}i)e^{i\pi} = -\sqrt{3}i$$

$$z_4 = 1 + (1 + \sqrt{3}i)e^{-i\pi/2} = (1 + \sqrt{3}) - i$$



29. Let $|z| = 2$ and $w = \frac{z+1}{z-1}$ where $z, w \in C$ (where C is the set of complex numbers). If M and m respectively be the greatest and least modulus of w , then find the value of $(2010m + M)$.

माना $|z| = 2$ और $w = \frac{z+1}{z-1}$ जहाँ $z, w \in C$ (जहाँ C सम्मिश्र संख्याओं का समुच्चय है) यदि M और m क्रमशः w के अधिकतम और न्यूनतम मापांक हैं। तब $(2010m + M)$ का मान ज्ञात कीजिए।

Ans. 673

Sol. Let $z = a + ib$ hence $a, b \in [-2, 2]$

$$w = \frac{(a+1) + ib}{(a-1) + ib}; \quad a^2 + b^2 = 4$$

$$|w| = \sqrt{\frac{(a+1)^2 + b^2}{(a-1)^2 + b^2}} = \sqrt{\frac{a^2 + b^2 + 2a + 1}{a^2 + b^2 - 2a + 1}} = \sqrt{\frac{5 + 2a}{5 - 2a}}$$

$$|w|_{\max} = \sqrt{\frac{5+4}{1}} = 3 = M, \quad \text{when } a = 2 \quad (z = 2)$$

$$|w|_{\min} = \sqrt{\frac{5-4}{9}} = \frac{1}{3} = m, \quad \text{when } a = -2 \quad (z = -2)$$

Hence $(2010m + M) = 673$ Ans.

30. If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then the minimum value of $|2z - 6 + 5i|$ is _____. [JEE-2011]

यदि z एक सम्मिश्र संख्या है जो $|z - 3 - 2i| \leq 2$, को सन्तुष्ट करती है तो $|2z - 6 + 5i|$ का न्यूनतम मान होगा _____ [JEE-2011]

Ans. 5

Sol. Given, $|z - 3 - 2i| \leq 2$ _____ (1)

To find minimum of $|2z - 6 + 5i|$

$$\text{Or } 2 \left| z - 3 + \frac{5}{2}i \right|,$$

using triangle inequality

$$\text{i.e., } ||z_1| - |z_2|| \leq |z_1 + \bar{z}_2|$$

$$\therefore \left| z - 3 + \frac{5}{2}i \right|$$

$$= \left| z - 3 - 2i + 2i + \frac{5}{2}i \right| = \left| (z - 3 - 2i) + \frac{9}{2}i \right| \geq \left| z - 3 - 2i \right| - \frac{9}{2} \geq \left| 2 - \frac{9}{2} \right| \geq \frac{5}{2}$$

$$\Rightarrow \left| z - 3 + \frac{5}{2}i \right| \geq \frac{5}{2} \text{ or } |2z - 6 + 5i| \geq 5$$

DE MOIVRE'S THEOREM

31. If Z is a non-real complex number, then find the minimum value of $\frac{\text{Im } Z^5}{\text{Im}^5 Z}$.

यदि Z एक अवास्तविक सम्मिश्र संख्या है, तो $\frac{\text{Im } Z^5}{\text{Im}^5 Z}$ का न्यूनतम मान ज्ञात करें।

Ans. -4

Sol. Let $z = re^{i\theta}$

$$\frac{\text{Im } z^5}{(\text{Im } z)^5} = \frac{r^5 \sin 5\theta}{r^5 \sin^5 \theta}$$

$$\text{say, } f(\theta) = \frac{\sin 5\theta}{\sin^5 \theta}$$

equatinig first derivative to 0

$$f'(\theta) = 0$$

$$\Rightarrow \sin^5 \theta \cdot 5 \cos 5\theta - \sin 5\theta \cdot 5 \sin^4 \theta \cdot \cos \theta$$

$$\Rightarrow \sin \theta \cdot \cos 5\theta = \cos \theta \cdot \sin 5\theta$$

$$\Rightarrow \tan \theta = \tan 5\theta$$

$$\Rightarrow \theta = 0, \frac{\pi}{4} \dots$$

but $f(0)$ is indeterminate

$$f\left(\frac{\pi}{4}\right) = -4$$

$$\therefore \text{Minimum value of } \frac{\operatorname{Im} z^5}{(\operatorname{Im} z)^5} \text{ is } -4$$

CUBE ROOT OF UNITY

32. Find the sum of the series $1(2-\omega)(2-\omega^2) + 2(3-\omega)(3-\omega^2) \dots (n-1)(n-\omega)(n-\omega^2)$ where ω is one of the imaginary cube root of unity.

श्रेणी $1(2-\omega)(2-\omega^2) + 2(3-\omega)(3-\omega^2) \dots (n-1)(n-\omega)(n-\omega^2)$ का योग ज्ञात कीजिये जहाँ ω इकाई का काल्पनिक घनमूल है।

Ans. $\left[\frac{n(n+1)}{2} \right]^2 - n$

Sol. $1(2-w)(2-w^2) + 2(3-w)(3-w^2) + (n-1)(n-w)(n-w^2)$

$$T_n = n(n+1-w)(n+1-w^2)$$

$$= (n^2 + n - nw)(n+1-w^2)$$

$$= n^3 + n^2 - n \cdot 2w^2 + n^3 + n - nw^2 - n^2w - nw + nw^3$$

$$= n^3 + n^2(2-w^2-w) + n(1-w-w^2+1)$$

$$= n^3 + 3n^2 + 3n$$

$$S_n = \sum T_n = \sum (n^3 + 3n^2 + 3n)$$

$$= \sum n^3 + 3\sum n^2 + 3\sum n$$

$$= \left[\frac{n(n+1)}{2} \right]^2 + \frac{3n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2}$$

$$= \frac{n(n+1)}{2} \left[\frac{n(n+1)}{2} + (2n+1) + 3 \right]$$

$$= \frac{n(n+1)}{4} [n^2 + n + 6 + 4n + 2]$$

$$S_n = \sum T_n = \frac{n(n+1)}{4} [n^2 + 5n + 8]$$

$$S_{n-1} = \frac{(n-1)n}{4} [n^2 + 1 - 2n + 5n - 5 + 8]$$

$$= \frac{(n-1)n}{4} [n^2 + 3n + 4]$$

33. Let ω be the complex number $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex number z satisfying

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0 \text{ is equal to.} \quad [\text{JEE-2010}]$$

माना ω एक सम्मिश्र संख्या $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$ है। तो विभिन्न सम्मिश्र संख्यायें z जो $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ को संतुष्ट

करती है, की संख्या है।

[JEE-2010]

Ans. 1

Sol. $\omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\begin{vmatrix} z & \omega & \omega^2 \\ z & z+\omega^2 & 1 \\ z & 1 & z+\omega \end{vmatrix} = 0 \quad \Rightarrow \quad \begin{vmatrix} 1 & \omega & \omega^2 \\ 1 & z+\omega^2 & 1 \\ 1 & 1 & z+\omega \end{vmatrix} = 0$$

$$z = 0$$

$$\begin{vmatrix} 1 & \omega & \omega^2 \\ 0 & z+\omega^2-\omega & 1-\omega^2 \\ 0 & 1-\omega & z+\omega-\omega^2 \end{vmatrix} = 0$$

$$(z + \omega^2 - \omega)(z + \omega - \omega^2) - (1 - \omega)(1 - \omega^2) = 0$$

$$z^2 = 0$$

only one solution

n^{th} ROOT OF UNITY

34. If $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -3/2$ then prove that :

यदि $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -3/2$ तो सिद्ध कीजिये :

(a) $\Sigma \cos 2\alpha = 0 = \Sigma \sin 2\alpha$

(b) $\Sigma \sin(\alpha + \beta) = 0 = \Sigma \cos(\alpha + \beta)$

(c) $\Sigma \sin^2 \alpha = \Sigma \cos^2 \alpha = 3/2$

(d) $\Sigma \sin 3\alpha = 3 \sin(\alpha + \beta + \gamma)$

(e) $\Sigma \cos 3\alpha = 3 \cos(\alpha + \beta + \gamma)$

Ans.

Sol. Given $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -3/2$

$$2 \cos(\alpha - \beta) + 2 \cos(\beta - \alpha) + 2 \cos(\gamma - \alpha) = -3$$

$$2 \cos(\alpha - \beta) + 2 \cos(\beta - \gamma) + 2 \cos(\gamma - \alpha) + 3 = 0$$

$$[2 \cos \alpha \cos \beta + 2 \sin \alpha \sin \beta] + [2 \cos \beta \cos \gamma + 2 \sin \beta \sin \gamma] + [2 \cos \gamma \cos \alpha + 2 \sin \gamma \sin \alpha] + 3 = 0$$

$$[2 \cos \alpha \cos \beta + 2 \cos \beta \cos \gamma + 2 \cos \gamma \cos \alpha] + [2 \sin \alpha \sin \beta + 2 \sin \beta \sin \gamma + 2 \sin \gamma \sin \alpha] + (\sin^2 \alpha + \cos^2 \alpha) + (\sin^2 \beta + \cos^2 \beta) + (\sin^2 \gamma + \cos^2 \gamma) = 0$$

$$\Rightarrow (\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + 2 \cos \alpha \cos \beta + 2 \cos \beta \cos \gamma + 2 \cos \gamma \cos \alpha) + (\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + 2 \sin \alpha \sin \beta + 2 \sin \beta \sin \gamma + 2 \sin \gamma \sin \alpha) = 0$$

$$(\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0$$

$$= (\cos \alpha + \cos \beta + \cos \gamma) = 0 \text{ and } \sin \alpha + \sin \beta + \sin \gamma = 0$$

35. Resolve $Z^5 + 1$ into linear & quadratic factors with real coefficients. Deduce that : $4 \cdot \sin \frac{\pi}{10} \cdot \cos \frac{\pi}{5} = 1$.

$Z^5 + 1$ को रेखीय तथा द्विघातीय गुणनखण्डों के रूप में व्यक्त कीजिये जिनके गुणांक वास्तविक हैं। दर्शाइये : $4 \cdot \sin \frac{\pi}{10} \cdot \cos \frac{\pi}{5} = 1$.

Ans. $(Z + 1)(Z^2 - 2Z \cos 36^\circ + 1)(Z^2 - 2Z \cos 108^\circ + 1)$

Sol. $Z^5 + 1 = 0 \Rightarrow Z^5 = -1$

$$= \cos(2n\pi + \pi) + i \sin(2n\pi + \pi)$$

$$= e^{i(2n\pi + \pi)} = e^{i(2n+1)\pi}$$

$$n = 0, 1, 2, 3, 4$$

$$n = 0 \Rightarrow Z = e^{i\frac{\pi}{5}}$$

$$n = 1 \Rightarrow z = e^{i\frac{3\pi}{5}}$$

$$n = 2 \quad Z = e^{i\pi}$$

$$n = 3 \Rightarrow Z = e^{i\frac{7\pi}{5}} = e^{-i\frac{\pi}{5}}$$

$$\begin{aligned} \left(Z - e^{i\frac{3\pi}{5}} \right) \left(Z - e^{-i\frac{3\pi}{5}} \right) &= Z^2 + 1 - Z \left(e^{i\frac{3\pi}{5}} + e^{-i\frac{3\pi}{5}} \right) \\ &= Z^2 - 2Z \cos \frac{3\pi}{5} + 1 \end{aligned}$$

$$Z^5 + 1 = (z + 1) \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right) \left(Z^2 - 2Z \cos \frac{\pi}{5} + 1 \right)$$

$$4 \sin \frac{\pi}{10} \cos \frac{\pi}{5} = 4(0.309016)(0.809016) = 1$$

Paragraph for Question Nos. 36 to 38

For the complex number $w = \frac{4z - 5i}{2z + 1}$

सम्मिश्र संख्या $w = \frac{4z - 5i}{2z + 1}$ के लिए

36. The locus of z , when w is a real number other than 2, is :

(A) a point circle

(B) a straight line with slope $-\frac{5}{2}$ and y-intercept $\frac{5}{4}$

(C) a straight line with slope $\frac{5}{2}$ and y-intercept $\frac{5}{4}$

(D) a straight line passing through the origin

z का बिन्दु पथ, जबकि w , 2 को छोड़ कर कोई भी वास्तविक संख्या है :

(A) एक बिन्दु वृत्त

(B) $-\frac{5}{2}$ प्रवणता तथा $\frac{5}{4}$, y अन्तःखण्ड वाली सरल रेखा

(C) $\frac{5}{2}$ प्रवणता तथा $\frac{5}{4}$, y अन्तःखण्ड वाली सरल रेखा

(D) मूल बिन्दु से जाने वाली सरल रेखा

Ans. C

Sol. (i) $w = \frac{4z - 5i}{2z + 1}$ ($w \neq z$)

if w is real then $w = \bar{w}$

$$\bar{w} = \frac{4\bar{z} + 5i}{\underbrace{2\bar{z} + 1}_{\bar{w}}} = \frac{4z - 5i}{\underbrace{2z + 1}_w}$$

$$(4\bar{z} + 5i)(2z + 1) = (4z - 5i)(2\bar{z} + 1)$$

$$8z\bar{z} + 4\bar{z} + 10zi + 5i = 8z\bar{z} + 4z - 10\bar{z}i - 5i$$

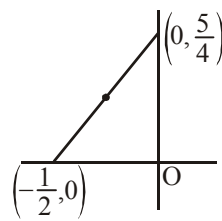
$$4(z - \bar{z}) - 10i(z + \bar{z}) - 10i = 0$$

$$8iy - 20ix - 10i = 0$$

$$4y - 10x - 5 = 0 \quad \Rightarrow \quad 10x - 4y + 5 = 0$$

Alternatively:

$$w = \frac{2\left(z - \frac{5}{4}\right)}{\left(z + \frac{1}{2}\right)} \quad \text{given } \frac{z - \frac{5}{4}}{z + \frac{1}{2}} \text{ is root } \left(-\frac{1}{2}, 0\right)$$



$\Rightarrow z$ moves on line formed $\left(-\frac{1}{2}, 0\right)$ and $\left(0, \frac{5}{4}\right)$ i.e., $m = \frac{5}{2}$ and y-intercept $\frac{5}{4}$.]

37. The locus of z , when w is a purely imaginary number is

(A) a circle with centre $\left(\frac{1}{2}, -\frac{5}{4}\right)$ passing through origin.

(B) a circle with centre $\left(-\frac{1}{4}, \frac{5}{8}\right)$ passing through origin.

(C) a circle with centre $\left(\frac{1}{4}, -\frac{5}{8}\right)$ and radius $\frac{\sqrt{29}}{8}$

(D) any other circle

z का बिन्दु पथ जबकि w पूर्णतः काल्पनिक संख्या है :

(A) $\left(\frac{1}{2}, -\frac{5}{4}\right)$ केन्द्र तथा $(0, 0)$ से जाने वाला वृत्त

(B) $\left(-\frac{1}{4}, \frac{5}{8}\right)$ केन्द्र तथा $(0, 0)$ से जाने वाला वृत्त

(C) केन्द्र $\left(\frac{1}{4}, -\frac{5}{8}\right)$ और त्रिज्या $\frac{\sqrt{29}}{8}$ का वृत्त

(D) कोई अन्य वृत्त

Ans. B

Sol. (ii) If w is purely imaginary then

$$w + \bar{w} = 0$$

$$\frac{4z - 5i}{2z + 1} + \frac{4\bar{z} + 5i}{2\bar{z} + 1} = 0$$

$$(4\bar{z} + 5i)(2z + 1) + (4z - 5i)(2\bar{z} + 1) = 0$$

simplifying $16z\bar{z} + 4(z + \bar{z}) + 10i(z - \bar{z}) = 0$

$$16(x^2 + y^2) + 8x - 20y = 0$$

$$x^2 + y^2 + \frac{x}{2} - \frac{5}{4}y = 0$$

Alternatively: When w is purely imaginary i.e., $\frac{x - \frac{5}{4}}{z + \frac{1}{2}}$ is purely imaginary, hence the curve will be the circle

described on $\left(0, \frac{5}{4}\right)$ and $\left(-\frac{1}{2}, 0\right)$ as denotes is $x\left(x + \frac{1}{2}\right) + y\left(y - \frac{5}{4}\right) = 0$

which passes through origin.

38. The locus of z , when $|w| = 1$ is :

(A) a circle with centre $\left(-\frac{5}{8}, \frac{1}{4}\right)$ and radius $\frac{1}{2}$ (B) a circle with centre $\left(\frac{1}{4}, -\frac{5}{8}\right)$ and radius $\frac{1}{2}$

(C) a circle with centre $\left(\frac{5}{8}, -\frac{1}{4}\right)$ and radius $\frac{1}{2}$ (D) any other circle

z का बिन्दुपथ, यदि $|w| = 1$:

(A) केन्द्र $\left(-\frac{5}{8}, \frac{1}{4}\right)$ और त्रिज्या $\frac{1}{2}$ का वृत्त

(B) केन्द्र $\left(\frac{1}{4}, -\frac{5}{8}\right)$ और त्रिज्या $\frac{1}{2}$ का वृत्त

(C) केन्द्र $\left(\frac{5}{8}, -\frac{1}{4}\right)$ और त्रिज्या $\frac{1}{2}$ का वृत्त

(D) कोई अन्य वृत्त

Ans. D

Sol. $|w| = 1$

$$w\bar{w} = 1$$

$$\left(\frac{4z - si}{2z + 1}\right)\left(\frac{4\bar{z} + si}{2\bar{z} + 1}\right) = 1$$

$$16(z)^2 + 25 = 4(z)^2 + 2z + 2\bar{z} + 1$$

$$12(z)^2 - 2(z + \bar{z}) + 24 = 0$$

$$12(x^2 + y^2) - 2(2x) + 24 = 0$$

$$x^2 + y^2 - \frac{1}{3}x + 2 = 0$$

Paragraph for Question Nos. 39 to 41

Let A, B, C be three sets of complex numbers as defined below.

$$A = \{z : |z + 1| \leq 2 + \operatorname{Re}(z)\}, \quad B = \{z : |z - 1| \geq 1\} \text{ and } C = \left\{z : \left|\frac{z-1}{z+1}\right| \geq 1\right\}$$

माना A, B, C तीन सम्मिश्र संख्याएँ इस प्रकार परिभाषित है कि

$$A = \{z : |z + 1| \leq 2 + \operatorname{Re}(z)\}, \quad B = \{z : |z - 1| \geq 1\} \text{ तथा } C = \left\{z : \left|\frac{z-1}{z+1}\right| \geq 1\right\}$$

39. The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is :

क्षेत्र $A \cap B \cap C$ में बिन्दुओं की संख्या जिनके निर्देशांक पूर्णांक है :

(A) 4 (B) 5 (C) 6 (D) 10

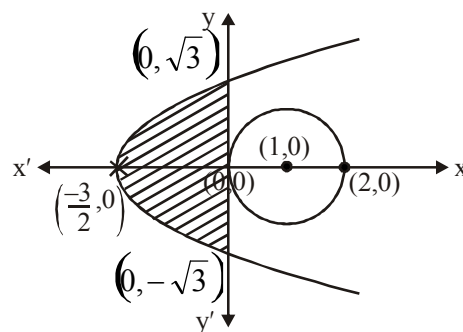
Ans. B

Sol. For A, $|z + 1| \leq 2 + \operatorname{Re}(z)$
 $\Rightarrow (x + 1)^2 + y^2 \leq 4 + 4x + x^2$
 $\Rightarrow y^2 \leq 3 + 2x$

$$\Rightarrow y^2 \leq 2\left(x + \frac{3}{2}\right) \quad \dots(1)$$

For B, $|z - 1| \geq 1$
 $\Rightarrow (x - 1)^2 + y^2 \geq 1 \quad \dots(2)$

For C, $|z - 1|^2 \geq |z + 1|^2$
 $\Rightarrow (z - 1)(\bar{z} - 1) \geq (z + 1)(\bar{z} + 1)$



$$\Rightarrow (z\bar{z} - \bar{z} - z + 1) \geq (z\bar{z} + \bar{z} + z + 1)$$

$$\Rightarrow z + \bar{z} \leq 0$$

$$\text{i.e. } x \leq 0 \quad \dots(3)$$

- (i) $(-1,0), (-1,1), (-1,-1), (0,0), (0,1), (0,-1)$ but $z = -1$ is not in the domain in set C
 $\therefore$ Total number of point(s) having integral coordinates in the region $A \cap B \cap C$ is 5.

40. The area of region bounded by $A \cap B \cap C$ is :

$A \cap B \cap C$ से परिबद्ध क्षेत्र का क्षेत्रफल :

- (A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $4\sqrt{3}$ (D) 2

Ans. A

Sol. (ii) Required area = $2 \int_{-\frac{3}{2}}^0 \sqrt{2\left(x + \frac{3}{2}\right)} dx = 2\sqrt{3}$ (square units)

41. The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude is :

क्षेत्र $A \cap B \cap C$ में सम्मिश्र संख्या का वास्तविक भाग, जिसका कोणांक का मुख्य मान अधिकतम है :

- (A) -1 (B) $-\frac{3}{2}$ (C) $\frac{1}{2}$ (D) -2

Ans. B

Sol. (iii) Clearly $z = \frac{-3}{2} + i0$ is the complex number in the region $A \cap B \cap C$ and having maximum amplitude.

$$\therefore \text{Re}(z) = \frac{-3}{2}$$

42. If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$, then the value of 'r' can be :

यदि $(1 + ir)^3$ को $s(1 + i)$ के रूप में लिखा जाता है कुछ वास्तविक 's' के लिए जहां 'r' भी वास्तविक है और $i = \sqrt{-1}$ तब 'r' का मान हो सकता है :

- (A) $\cot \frac{\pi}{8}$ (B) $\sec \pi$ (C) $\tan \frac{\pi}{12}$ (D) $\tan \frac{5\pi}{12}$

Ans. BCD

Sol. $(1 + ir)^3 = 1^3 + 3ir + 3(ir)^2 + (ir)^3$

$$= 1 + i(3r) + 3(-1)r^2 - r^3i = s + si$$

$$= (1 - 3r^2) + i(3r - r^3) = s + si$$

$$= 1 - 3r^2 = s \text{ and } 3r - r^3 = s$$

$$1 - 3r^2 = 3r - r^3$$

$$1 = \frac{3r - r^3}{1 - 3r^2}$$

$$1 = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = \tan 3\theta$$

$$3\theta = \frac{\pi}{4}, \quad r = \tan \theta$$

$$\theta = \frac{\pi}{12}, \quad r = \tan \frac{\pi}{12}$$

ONE OR MORE THAN ONE CORRECT TYPE

43. Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t)z_1 + tz_2$ for some real number t with $0 < t < 1$. If $\text{Arg}(w)$ denotes the principal argument of a nonzero complex number w , then : **[JEE-2010]**

माना z_1 तथा z_2 दो भिन्न सम्मिश्र संख्यायें हैं तथा $z = (1 - t)z_1 + tz_2$ है जहाँ t कोई वास्तविक संख्या है जिसमें $0 < t < 1$ है। यदि $\text{Arg}(w)$ का अशून्य सम्मिश्र संख्या w का मुख्य कोण दर्शाता है, तो : **[JEE-2010]**

(A) $|z - z_1| + |z - z_2| = |z_1 - z_2|$ (B) $\text{Arg}(z - z_1) = \text{Arg}(z - z_2)$

(C) $\begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 \\ z_2 - z_1 & \bar{z}_2 - \bar{z}_1 \end{vmatrix} = 0$ (D) $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$

Ans. ACD

Sol. Given, $z = \frac{(1 - t)z_1 + tz_2}{(1 - t) + 1}$

$$\begin{array}{ccc} \text{A} & \text{P} & \text{B} \\ | & | & | \\ z_1 & z & z_2 \end{array}$$

$$t : (1 - t)$$

Clearly, z divides z_1 and z_2 in the ratio of $t : (1 - t)$, $0 < t < 1$

$$\Rightarrow AP + BP = AB$$

$$\text{ie, } \Rightarrow |z - z_1| + |z - z_2| = |z_1 - z_2|$$

$\Rightarrow$ Option (a) is true.

$$\text{And } \arg(z - z_1) = \arg(z_2 - z) = \arg(z_2 - z_1)$$

$\Rightarrow$ (b) is false and (d) is true.

$$\text{Also, } \arg(z - z_1) = \arg(z_2 - z_1)$$

$$\Rightarrow \arg\left(\frac{z - z_1}{z_2 - z_1}\right) = 0$$

$\therefore \frac{z - z_1}{z_2 - z_1}$ is purely real.

$$\Rightarrow \frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} \text{ or } \left| \frac{z - z_1}{z_2 - z_1} \cdot \frac{\bar{z}_2 - \bar{z}_1}{\bar{z} - \bar{z}_1} \right| = 0$$

$\therefore$ option c is correct.

44. POQ is a straight line through the origin O. P and Q represent the complex number $a + ib$ and $c + id$ respectively and $OP = OQ$. Then :

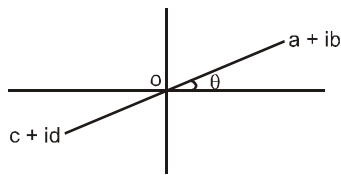
- (A) $|a + ib| = |c + id|$ (B) $a + c = b + d$
(C) $\arg(a + ib) = \arg(c + id)$ (D) None of these

मूलबिन्दु O से गुजरने वाली रेखा POQ है। P तथा Q सम्मिश्र संख्याओं $a + ib$, $c + id$ को प्रदर्शित करते हैं तथा $OP = OQ$ है। तब :

- (A) $|a + ib| = |c + id|$ (B) $a + c = b + d$
(C) कोणांक $(a + ib) =$ कोणांक $(c + id)$ (D) इनमें से कोई नहीं

Ans. AB

Sol.



$$|a + ib| = |(c + id)|$$

$$a + c = b + d$$

Alter :

It is given that origin is the mid point of the line POQ as $OP = OQ$.

So, coordinates of origin will be $\frac{(a + c)}{2} + \frac{(b + d)}{2}i$, but coordinates of the origin are given as $0 + 0i$.

On comparing we get,

$$\frac{a+c}{2} = 0 \quad \text{_____ (1) and } \frac{b+d}{2} = 0 \quad \text{_____ (2)}$$

Equation (1) and (2)

$$\Rightarrow \frac{a+c}{2} = \frac{b+d}{2}$$

$$\Rightarrow a+c = b+d$$

45. If z satisfies the inequality $|z - 1 - 2i| \leq 1$, then :

(A) $\min(\arg(z)) = \tan^{-1}\left(\frac{3}{4}\right)$

(B) $\max(\arg(z)) = \frac{\pi}{2}$

(C) $\min(|z|) = \sqrt{5} - 1$

(D) $\max(|z|) = \sqrt{5} + 1$

यदि z असमिका $|z - 1 - 2i| \leq 1$ को संतुष्ट करता हो, तो :

(A) न्यूनतम (कोणांक (z)) = $\tan^{-1}\left(\frac{3}{4}\right)$

(B) अधिकतम (कोणांक (z)) = $\frac{\pi}{2}$

(C) न्यूनतम ($|z|$) = $\sqrt{5} - 1$

(D) अधिकतम ($|z|$) = $\sqrt{5} + 1$

Ans. ABCD

Sol. $\max |z| = d + r = \sqrt{5} + 1$

$\min |z| = d - r = \sqrt{5} - 1$

$d = OC = \sqrt{5}$

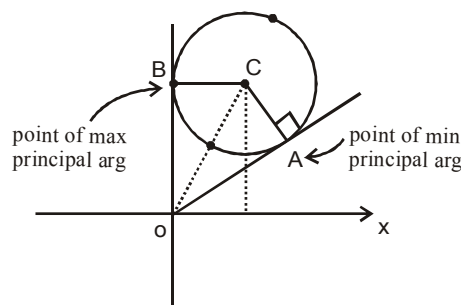
$r = 1$

$\theta = \angle OCX = \tan^{-1} \frac{2}{1}$

$\alpha = \angle OCA = \tan^{-1} \frac{1}{2} \quad \left(\because \sin \alpha = \frac{1}{\sqrt{5}} \right)$

So principal Arg of A = $\theta - \alpha = \tan^{-1} 2 - \tan^{-1} \frac{1}{2}$

$= \tan^{-1} \frac{2 - \frac{1}{2}}{1 + 1} = \tan^{-1} \frac{3}{4}$



46. If $|z_1| = |z_2| = |z_3| = 1$ and z_1, z_2, z_3 are represented by the vertices of an equilateral triangle then :

(A) $z_1 + z_2 + z_3 = 0$

(B) $z_1 z_2 z_3 = 1$

(C) $z_1 z_2 + z_2 z_3 + z_3 z_1 = 0$

(D) None of these

यदि $|z_1| = |z_2| = |z_3| = 1$ हो तथा एक समबाहु त्रिभुज के शीर्ष z_1, z_2, z_3 हो, तो :

- (A) $z_1 + z_2 + z_3 = 0$ (B) $z_1 z_2 z_3 = 1$ (C) $z_1 z_2 + z_2 z_3 + z_3 z_1 = 0$ (D) इनमें से कोई नहीं

Ans. AC

Sol. All the three vertices lies on circle $|z| = 1$

take $z_1 = z_1, z_2 = z_1 \omega, z_3 = z_1 \omega^2$

so $z_1 + z_2 + z_3 = z_1(1 + \omega + \omega^2) = 0$

$z_1 z_2 z_3 = z_1^3 \dots (i)$

$z_1 z_2 + z_2 z_3 + z_3 z_1 = z_1^2 (\omega + \omega^3 + \omega^2) = 0$

if $z_1 + z_2 + z_3 = 0$ then $z_1^3 + z_2^3 + z_3^3 = 3z_1 z_2 z_3$

from (i) $z_1^3 + z_2^3 + z_3^3 = 3z_1^3$

$z_2^3 + z_3^3 = 2z_1^3$

hence proved.

47. Let A and B be two distinct points denoting the complex numbers α and β respectively. A complex number z lies between A and B where $z \neq \alpha, z \neq \beta$. Which of the following relation(s) hold good?

(A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$

(B) $\exists$ a positive real number 't' such that $z = (1 - t)\alpha + t\beta$

(C) $\begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$

(D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

माना A और B दो भिन्न बिन्दु सम्मिश्र संख्या α और β को दर्शाते हैं। सम्मिश्र संख्या z , A और B के मध्य है जहाँ $z \neq \alpha, z \neq \beta$. तब सत्य कथन होंगे :

(A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$

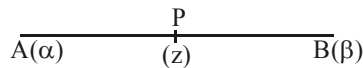
(B) $\exists$ एक धनात्मक वास्तविक संख्या 't' इस प्रकार की $z = (1 - t)\alpha + t\beta$

(C) $\begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$

(D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

Ans. ABCD

Sol. $AP + PB = AB$



$|z - \alpha| + |\beta - z| = |\beta - \alpha| \Rightarrow$ A is True

Now $z = \alpha + t(\beta - \alpha)$

$= (1 - t)\alpha + t\beta$ where $t \in (0, 1) \Rightarrow$ B is True

again $\frac{z - \alpha}{\beta - \alpha}$ is real $\Rightarrow \frac{z - \alpha}{\beta - \alpha} = \frac{\bar{z} - \bar{\alpha}}{\bar{\beta} - \bar{\alpha}}$

$\Rightarrow \begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$ Ans.

$$\text{also } \begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0 \text{ if and only if } \begin{vmatrix} z-\alpha & \bar{z}-\bar{\alpha} & 0 \\ \alpha & \bar{\alpha} & 1 \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} & 0 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} (z-\alpha) & \bar{z}-\bar{\alpha} \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} \end{vmatrix} = 0 \text{ Ans.}$$

48. Let z_1, z_2, z_3 be non-zero complex numbers satisfying the equation $z^4 = iz$. Which of the following statement(s) is/are correct?

(A) The complex number having least positive argument is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

(B) $\sum_{k=1}^3 \text{Amp}(z_k) = \frac{\pi}{2}$

(C) Centroid of the triangle formed by z_1, z_2 and z_3 is $\left(\frac{1}{\sqrt{3}}, \frac{-1}{3}\right)$

(D) Area of triangle formed by z_1, z_2 and z_3 is $\frac{3\sqrt{3}}{2}$

माना z_1, z_2, z_3 अशून्य सम्मिश्र संख्याएँ हैं जो समीकरण $z^4 = iz$ को सन्तुष्ट करते हैं तब निम्न में से सत्य कथन होंगे?

(A) सम्मिश्र संख्या, जिसका न्यूनतम धनात्मक कोणांक है, $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ है

(B) $\sum_{k=1}^3 \text{Amp}(z_k) = \frac{\pi}{2}$

(C) z_1, z_2 और z_3 से निर्मित त्रिभुज का केन्द्रक $\left(\frac{1}{\sqrt{3}}, \frac{-1}{3}\right)$ है

(D) z_1, z_2 और z_3 द्वारा निर्मित त्रिभुज का क्षेत्रफल $\frac{3\sqrt{3}}{2}$ है

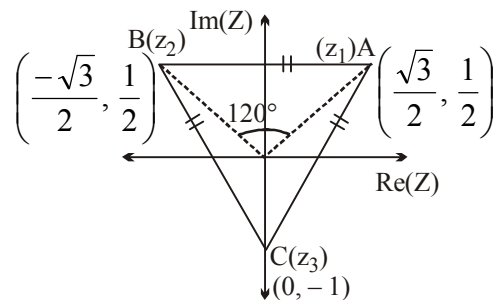
Ans. AB

Sol. We have $z^4 = iz \Rightarrow z^3 = i$

$$\Rightarrow z = e^{i(4k+1)\frac{\pi}{6}} \text{ (Using D.M.T.)}$$

Put $k = 0, 1, 2$, we get

$$z_1 = e^{i\frac{\pi}{6}}, z_2 = e^{i\frac{5\pi}{6}} \text{ and } z_3 = e^{i\frac{3\pi}{2}}$$



Clearly triangle formed by z_1, z_2 and z_3 is equilateral.

$\therefore$ centroid of ΔABC is $(0, 0)$ and Area $(\Delta ABC) = \frac{3\sqrt{3}}{4}$

49. If $z \in \mathbb{C}$, which of the following relation(s) represents a circle on an Argand diagram?

यदि $z \in \mathbb{C}$, तब निम्न में से कौन कौन से सम्मिश्र तल में वृत्त को दर्शाते हैं?

(A) $|z - 1| + |z + 1| = 3$

(B) $(z - 3 + i)(\bar{z} - 3 - i) = 5$

(C) $3|z - 2 + i| = 7$

(D) $|z - 3| = 2$

Ans. BCD

Sol. (A) is obviously ellipse

(B) $(z - \alpha)(\bar{z} - \bar{\alpha}) = 5$ where $\alpha = 3 - i$; $\bar{\alpha} = 3 + i$

$|z - \alpha|^2 = 5 \Rightarrow |z - \alpha| = \sqrt{5}$ circle with centre $(3, -1)$ and radius $= \sqrt{5} \Rightarrow$ (B) is correct

(C) $|z - (2 - i)| = \frac{7}{3} \Rightarrow$ circle with centre $(2, -1)$ and radius $= \frac{7}{3} \Rightarrow$ (C) is correct

(D) $|z - 3| = 2 \Rightarrow$ circle with centre $(3, 0)$ and radius $= 2 \Rightarrow$ (D) is correct

50. Equation of a straight line on the complex plane passing through a point P denoting the complex number α and perpendicular to the vector $\overrightarrow{OP}$ where 'O' is the origin can be written as :

सम्मिश्र तल में P (α) से जाने वाली तथा सदिश $\overrightarrow{OP}$ (जहाँ 'O' मूल बिन्दु है) के लम्बवत सरल रेखा है :

(A) $\text{Im}\left(\frac{z - \alpha}{\alpha}\right) = 0$ (B) $\text{Re}\left(\frac{z - \alpha}{\alpha}\right) = 0$ (C) $\text{Re}(\bar{\alpha}z) = 0$ (D) $\bar{\alpha}z + \alpha\bar{z} - 2|\alpha|^2 = 0$

Ans. BD

Sol. Required line is passing through P(α) and parallel to the vector $\overrightarrow{OQ}$

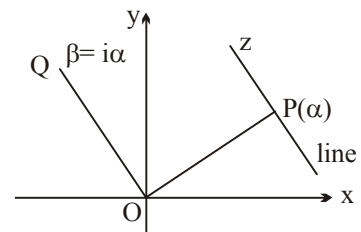
hence $z = \alpha + i\lambda\alpha, \lambda \in \mathbb{R}$

$\frac{z - \alpha}{\alpha} = \text{purely imaginary}$

$\Rightarrow \text{Re}\left(\frac{z - \alpha}{\alpha}\right) = 0 \Rightarrow$ (B) (multiply N^r and D^r by $\bar{\alpha}$)

$\Rightarrow \text{Re}((z - \alpha)\bar{\alpha}) = 0 \Rightarrow \text{Re}(z\bar{\alpha} - |\alpha|^2) = 0$

also $\frac{z - \alpha}{\alpha} + \frac{\bar{z} - \bar{\alpha}}{\bar{\alpha}} = 0$



$$\overline{\alpha}(z - \alpha) + \alpha(\bar{z} - \overline{\alpha}) = 0$$

$$\overline{\alpha} z + \alpha \bar{z} - 2|\alpha|^2 = 0 \Rightarrow \quad (D)$$

MATCH THE COLUMN

51. Match the statements in Column I with those in Column-II

[Note : Here z takes values in the complex plane and $\text{Im } z$ and $\text{Re } z$ denote, respectively, the imaginary part and the real part of z .] [JEE-2010]

Column I

(A) The set of points z satisfying

$$|z - i| |z| = |z + i| |z|$$

is contained in or equal to

(B) The set of points z satisfying

$$|z + 4| + |z - 4| = 10$$

is contained in or equal to

(C) If $|w| = 2$, then the set of points

$$z = w - \frac{1}{w} \text{ is contained in or equal to}$$

(D) If $|w| = 1$, then the set of points

$$z = w + \frac{1}{w} \text{ is contained in or equal to}$$

Column II

(p) an ellipse with eccentricity $\frac{4}{5}$

(q) the set of points z satisfying $\text{Im } z = 0$

(r) the set of points z satisfying $|\text{Im } z| \leq 1$

(s) the set of points z satisfying $|\text{Re } z| \leq 2$

(t) the set of points z satisfying $|z| \leq 3$

स्तम्भ-I का स्तम्भ-II से मिलान कीजिये।

[Note : यहाँ z सम्मिश्र तल में स्थित है तथा $\text{Im } z$ एवं $\text{Re } z$ क्रमशः z के काल्पनिक एवं वास्तविक भाग हैं]

[JEE-2010]

Column I

(A) $|z - i| |z| = |z + i| |z|$ को संतुष्ट करने

वाले बिन्दु के समुच्चय बराबर है या निहित है।

(B) $|z + 4| + |z - 4| = 10$ को संतुष्ट करने वाले बिन्दु किसमें निहित है या बराबर है

(C) यदि $|w| = 2$, तो समुच्चयों जिनके लिये

$$\text{बिन्दु } z = w - \frac{1}{w} \text{ में निहित या बराबर है}$$

(D) यदि $|w| = 1$, तो समुच्चयों जिनके लिये

$$\text{बिन्दु } z = w + \frac{1}{w} \text{ में निहित या बराबर है}$$

Column II

(p) एक दीर्घवृत्त जिसकी उत्केन्द्रता $\frac{4}{5}$

(q) $\text{Im } z = 0$ को संतुष्ट करने वाले बिन्दु z का समुच्चय

(r) $|\text{Im } z| \leq 1$ को संतुष्ट करने वाले बिन्दु z का समुच्चय

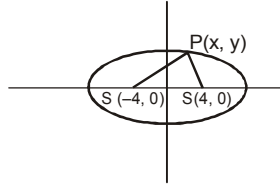
(s) $|\text{Re } z| \leq 2$ को संतुष्ट करने वाले बिन्दु z का समुच्चय

(t) $|z| \leq 3$ को संतुष्ट करने वाले बिन्दु z का समुच्चय

Ans. (A) q, r (B) p (C) p, s, t (D) q, r, s, t

Sol. $\Rightarrow |x + iy - i\sqrt{x^2 + y^2}| = |x + iy + i\sqrt{x^2 + y^2}| \Rightarrow x^2 + \left(y - \sqrt{x^2 + y^2}\right)^2 = x^2 + \left(y + \sqrt{x^2 + y^2}\right)^2$
 $\Rightarrow 4y\sqrt{x^2 + y^2} = 0 \Rightarrow y = 0 \Rightarrow \text{Im } z = 0$

(B) $|z + 4| + |z - 4| = 10$
 Ellipse with $2a = 10 \Rightarrow a = 5$
 $ae = 4 \Rightarrow e = \frac{4}{5}$



(C) Let $w = 2(\cos\theta + i\sin\theta)$

$$z = 2(\cos\theta + i\sin\theta) - \frac{(\cos\theta - i\sin\theta)}{2}$$

$$= \frac{3\cos\theta + 5i\sin\theta}{2} \Rightarrow x = \frac{3\cos\theta}{2}, y = \frac{5\sin\theta}{2}$$

$$= \frac{x^2}{9/4} + \frac{y^2}{25/4} = 1 \quad e = \frac{4}{5}$$

$$|z| = \sqrt{\frac{9\cos^2\theta}{4} + \frac{25\sin^2\theta}{4}} = \sqrt{\frac{9 + 16\sin^2\theta}{4}} = \sqrt{\frac{9}{4} + 4\sin^2\theta} \leq \frac{5}{2}$$

$$|\text{Re } z| = \left| \frac{3}{2}\cos\theta \right| \leq \frac{3}{2}$$

(D) $z = \cos\theta + i\sin\theta + \cos\theta - i\sin\theta = 2\cos\theta$
 $\therefore |z| \leq 2$
 $\therefore \text{Im}(z) = 0$
 $(\text{Re } z) \Rightarrow |2\cos\theta| \leq 2$
 $|z| \leq 2$

52. Match the following columns :

Column I

- (A) Let ω be a non real cube root of unity then the number of distinct elements in the set $\{(1 + \omega + \omega^2 + \dots + \omega^n)^m \mid m, n \in \mathbb{N}\}$ is
- (B) Let $1, \omega, \omega^2$ be the cube root of unity. The least possible degree of a polynomial with real coefficients having roots $2\omega, (2 + 3\omega), (2 + 3\omega^2), (2 - \omega - \omega^2)$, is

Column II

- (P) 4
- (Q) 5

(C) $\alpha = 6 + 4i$ and $\beta = (2 + 4i)$ are two complex numbers on the (R) 6

complex plane. A complex number z satisfying $\arg\left(\frac{z - \alpha}{z - \beta}\right) = \frac{\pi}{6}$ (S) 7

moves on the major segment of a circle whose radius is

निम्न को सुमेलित कीजिए :

Column I

Column II

(A) माना ω एक अवास्तविक इकाई का घनमूल है तो समुच्चय

(P) 4

$\{(1 + \omega + \omega^2 + \dots + \omega^n)^m \mid m, n \in \mathbb{N}\}$ में स्थित भिन्न अवयवों की संख्या होगी

(B) माना $1, \omega, \omega^2$ इकाई के घनमूल हैं। बहुपद का न्यूनतम सम्भावित घात जिसके मूल $2\omega, (2 + 3\omega), (2 + 3\omega^2), (2 - \omega - \omega^2)$ हैं तथा गुणांक वास्तविक हैं। (Q) 5

(C) $\alpha = 6 + 4i$ तथा $\beta = (2 + 4i)$ दो सम्मिश्र संख्याएँ सम्मिश्र तल में स्थित हैं (R) 6

एक सम्मिश्र संख्या z जो $\arg\left(\frac{z - \alpha}{z - \beta}\right) = \frac{\pi}{6}$ को संतुष्ट करती है वृत्त की (S) 7

दीर्घखण्ड में गमन कर रही है जिसकी त्रिज्या है

Ans. (A) S (B) Q (C) P

Sol.

53. Let z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$. Match the following :

Column I

Column II

(A) Maximum value of $|z_1 + z_2|$ (p) 3

(B) Minimum value of $|z_1 - z_2|$ (q) 1

(C) Minimum value of $|2z_1 + 3z_2|$ (r) 4

(D) Maximum value of $|z_1 - 2z_2|$ (s) 5

यदि $z_1, |z| = 1$ पर स्थित है तथा $z_2, |z| = 2$ पर स्थित है :

Column I

Column II

(A) $|z_1 + z_2|$ का अधिकतम मान (p) 3

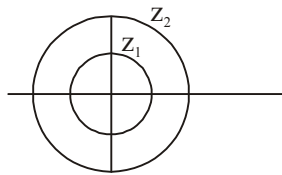
(B) $|z_1 - z_2|$ का न्यूनतम मान (q) 1

(C) $|2z_1 + 3z_2|$ का न्यूनतम मान (r) 4

(D) $|z_1 - 2z_2|$ का अधिकतम मान (s) 5

Ans. (A) p (B) q (C) r (D) s

Sol.



- (A) $|z_1 + z_2| \leq |z_1| + |z_2| \leq 2 + 1 \leq 3$
 (A) $\rightarrow P$
- (B) $|z_1 - z_2| \Rightarrow$ minimum distance b/w z_1 & $z_2 = 1$
 $|z_1 - z_2| \Rightarrow z_1$ तथा z_2 के मध्य दूरी = 1
 B $\rightarrow q$
- (C) $|2z_1 + 3z_2|$ minimum is न्यूनतम = $6 - 2 = 4$
 (C) $\rightarrow r$
- (D) $|z_1 - 2z_2| \leq |z_1| + |-2z_2|$
 $1 + 4 \leq 5$
 (D) $\rightarrow s$